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A critical systematic review of ensemble learning methods for predictive healthcare

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Abstract

Ensemble learning has emerged as a significant paradigm in predictive healthcare, offering enhanced accuracy, robustness, and generalization relative to single-model methods. The systematic review critically examines current advancements in ensemble-based illness prediction across five major disease categories: heart disease, diabetes, liver disease, kidney disease, and skin disease. Applying the PRISMA method, 287 studies published from 2020 to 2024 were found in major scientific databases, establishing the first group. Subsequent to the comprehensive screening and eligibility assessment, 91 excellent papers were selected for careful examination. Statistical analysis of these studies reveals that approximately 40% employ stacking or hybrid ensemble models with feature selection. These models consistently demonstrate the highest performance, achieving accuracy in the range of 94–97%. 20% of studies are based on simple Bagging or Voting based ensemble methods. In contrast, about 20% of the studies adopt bagging or voting-based ensemble methods, which generally result in comparatively lower accuracy. Deep learning–based ensemble approaches account for approximately 12% of the reviewed studies. Overall, this review provides a comprehensive quantitative assessment of ensemble learning techniques in predictive healthcare. It highlights dominant methodological trends, comparative performance outcomes, and various practical challenges related to computational overhead, limited explainability, and dataset biasness.

Keywords Systematic review, Healthcare, Disease prediction, PRISMA, Machine learning, Deep learning, Ensemble learning

1 Introduction

Predictive modeling in healthcare has dramatically changed the early diagnosis and management of diseases, thus leading to better patient outcomes and reduced costs for healthcare. The use of machine learning (ML) [1–4, 5, 6] and artificial intelligence (AI) [7, 8, 9–14] allows predictive models to analyze complex datasets, identify patterns, and forecast potential health risks. Heart disease remains one of the top causes of death worldwide. As per World Health Organization (WHO) [15] reports, approximately 17.9 million people die from cardiovascular diseases (CVDs) each year [16, 17], amounting to 31% of all global deaths shown in Fig. 1. It represented last five years death due to the major diseases.

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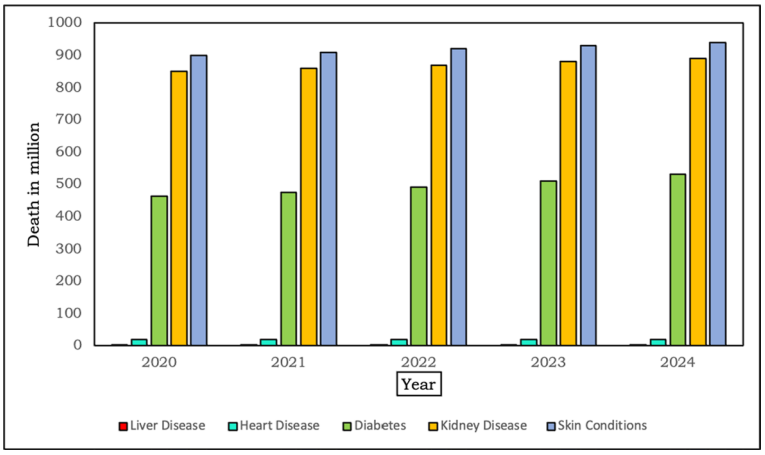


Fig. 1 last 5 years death in major diseases as per WHO [15]

Predictive models for the early detection of heart disease could greatly reduce the mortality rate. Diabetes is a chronic condition found in more than 537 million adults worldwide and severely threatens the life of several organs if not properly managed [18]. Predictive modeling has been used widely to identify high-risk individuals who may develop Type 2 diabetes. Liver diseases, such as hepatitis, fatty liver, and cirrhosis, account for a large number of morbidity and mortality worldwide. According to the Global Burden of Disease Study [19], liver diseases account for more than 2 million deaths per year. Chronic kidney disease (CKD) affects approximately 10% of the global population, with millions remaining undiagnosed due to the disease's asymptomatic nature in early stages. Skin diseases encompass a wide range of conditions, from benign disorders such as acne to life-threatening melanomas. According to the American Academy of Dermatology, skin cancer is the most common cancer in the United States, with 1 in 5 Americans expected to develop it in their lifetime [20]. Among the myriad of applications in healthcare, disease prediction has garnered significant attention, particularly for chronic and critical conditions such as heart disease, diabetes, liver disorders, kidney dysfunction, and skin diseases. Each of these conditions is a huge burden to global health, and the potential to predict their onset or progression can transform healthcare delivery.

The ML is a key and important part of AI [21] that can be used in many areas including healthcare because the world is full of different types of data [22, 23]. A wide variety of learning algorithms and methodologies exists [24, 25, 26], each assessed according to criteria such as efficacy and scalability. Three methodologies ML, deep learning (DL) [27–29] and ensemble learning [30, 31, 32] have been prominent in the AI domain.

The process of ensemble learning combines several basic models to make a single, stronger model that does better than each of its components [33]. This method reduces the possibility of overfitting through using the diversity provided by the baseline models. Ensemble learning has been effectively utilized in healthcare sector, constantly surpassing single-model methodologies [34–38]. Ensemble learning includes many strategies that differ in the training and integration of baseline models to improve overall performance. The most used techniques include averaging [39, 40], bagging [41, 42], random forests [41, 43], stacking [44, 45], and boosting [46], each providing unique approaches to enhance model accuracy and resilience.

Heart disease, diabetes, liver disorders, kidney disease and skin cancer are the major chronic diseases that still burden the world health by contributing to high morbidity, mortality and high healthcare expenditure globally. The increasing access to healthcare big data, in terms of scale, clinical, imaging and sensor data, has prompted increased use of ensemble learning methods when predicting early disease onset because they are more accurate and robust than single models. Nevertheless, some of these achievements fail to seal some important gaps. Ensemble models, and especially boosting- and deep learning-based ensembles, are not very interpretable, which limits their clinical trust and adoption. The bias of predictions in ensemble structures against class imbalance, which is typical in medical data where positive cases of a disease are few, persists. Also, the vast majority of current models are trained using static data and are not effective at handling dynamic patient data, including changing physiological measurements and longitudinal health history. It is important to tackle these issues with explainable ensemble models, imbalance-sensitive learning algorithms, and adaptive/online ensemble models in order to convert the body of ensemble learning research to reliable real-world clinical decision support systems.

The paper presents a comparative and critical review of ensemble-based predictive models for the early detection and management of heart disease, diabetes, liver disorders, kidney dysfunction, and skin diseases. By scrutinizing the applied methodologies, datasets, and performance evaluation measures, it presents a thorough comprehension of the ensemble modeling practice at present. Our review is the most extensive and current work in the area. We selected 287 articles published in the last five years, from 2020 to 2024. After a critical analysis, we finally included 91 articles from some of the leading publishers: Elsevier, IEEE, MDPI, Springer, among others. In contrast to the review by Mahajan et al., 2023 [47], our study encompasses a substantially greater number of articles: 45, featuring more current and diverse contributions. The review by Islam et al. [48] is limited to ML methodologies. Our research provides a comprehensive perspective by emphasising ensemble techniques, encompassing ML, DL, and hybrid methodologies. This unequivocally demonstrates that our review possesses greater breadth and profundity. This review also brings out challenges and gaps in literature and directions to future work by enhancing the efficacy and applicability of ensemble-based predictive healthcare systems.

1.1 Main contribution of the paper

Earlier machine learning disease prediction review research concentrated on single classifiers or typical machine learning algorithms and rarely touched ensemble learning. Several previous assessments only examined traditional ensemble approaches such bagging or random forests, not hybrid ensemble frameworks that incorporate various machine learning and deep learning models. Despite their rising use in complicated healthcare data circumstances like medical imaging and historical patient records, several evaluations ignored or barely mentioned ensemble deep learning algorithms like stacked convolutional or recurrent neural networks. Instead of comparing numerous illness categories, earlier studies focused on a single condition or dataset, limiting generalizability. Most surveys used narrative summaries without PRISMA-based screening, resulting in fragmented coverage and selection bias. Thus, a complete, up-to-date assessment of classical ensembles, hybrid models, and deep learning–based ensembles

in a single systematic framework was needed, especially for predictive healthcare applications from since 2020.

This review carefully examines all the methods used in ensemble machine learning methods in predicting diseases. There are several important things that affect how well ensemble methods work. These include baseline learning models, data sampling methods during training, varying baseline classifiers, and combining model results. The article analyses the pros and cons of each technique, giving useful information on their usefulness and limitations in real life. The main contribution of this paper as follow:

- The ensemble learning and its mathematical concept are well explained, providing a foundational understanding of how this approach addresses disease prediction issues.
- Introducing a PRISMA approach that includes additional screening steps to identify a sufficient number of the most trustworthy research on ensemble learning advancements.
- Provide an exhaustive review of the up-to-date research progress of ensemble learning in major disease prediction such as heart, diabets, kidney, liver and skin according to different performance metrics and trade-offs.
- Outline potential future directions for the development of the ensemble learning.

1.2 Organization of the paper

The structure of this survey is as follows:

Emerging trends in ensemble learning and challenges for disease prediction are presented in Sect. 2. Systematic review methodology and Research Questions are presented in Sect. 3. Section 4 shows key disease prediction research techniques based on ensemble learning from 2020 to 2024. This section also includes tabulated evaluation results and comparisons that align with the technical methodologies parameters. The main issues revealed in the survey are examined and discussed in Sect. 5. Then, a concise and perceptive conclusion summarising the paper's main conclusions and future directions is provided in Sect. 6. The concepts of the ensemble learning algorithm describe in Appendix in Section A.

2 Emerging trends in ensemble learning for disease prediction

Ensemble learning has become one of the strongest paradigms in disease prediction, utilizing the strengths of various models to improve predictive performance far beyond individual algorithms. Recent statistics reflect its importance in healthcare applications. As shown in Fig. 2, ensemble learning techniques focused on disease prediction and classification have been observed in a significant uptrend. In the year 2016, in total, there were 810 research articles; nil for dealt with disease prediction. But in 2024, this number surged to 9,430 research articles, including 125 for disease prediction. This growth reflects the increased popularity and adaptation among researchers into using techniques of ensemble learning to address critical challenges in the process of disease prediction and classification.

The increasing popularity results in a significant increase of research projects utilizing ensemble learning across several application domains. This section clarifies this tendency by examining data from the extensive database "Scopus" to assess the yearly rise of indexed publications on ensemble learning and its application domains from 2004

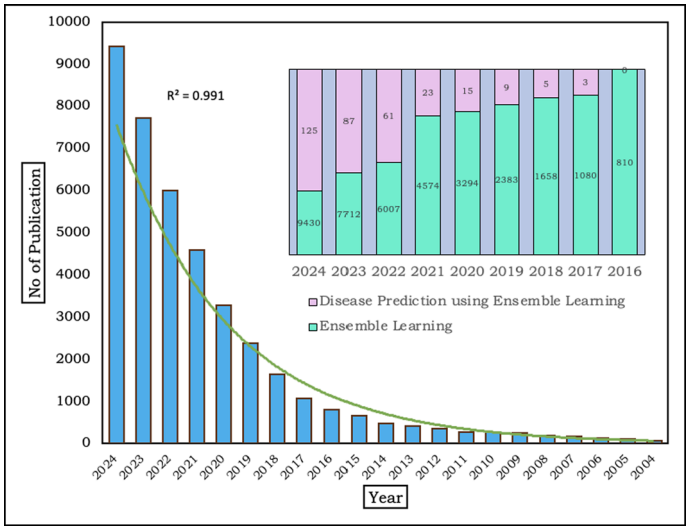


Fig. 2 Trends of ensemble learning from 2004 to 2024

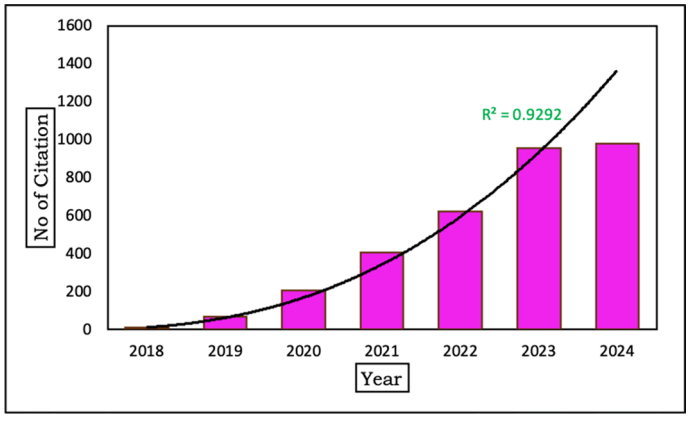


Fig. 3 Year wise growth of ensemble approaches from 2018 to 2024

to 2024. The study utilized keyword searches such as "Ensemble Learning", "Ensemble Machine Learning, "Ensemble Deep Learning" focusing on article titles, abstracts, and keywords. This analysis illustrates the increasing significance of ensemble approaches in ML research and their growing impact across several fields. Figure 2 depicts the yearly publication count of articles for the search term "Ensemble Learning" over the specified timeframe. The data indicates a total of 40,332 publications, demonstrating a consistent rise in the popularity and research involvement related to ensemble learning throughout time. The trend line, with a R^2 value of 0.991, signifies substantial and steady growth in ensemble learning research, underscoring its escalating relevance and extensive use across many domains. same os the grwoth of ensemble learning increase day by day shown in Fig. 3, Here in 2018 the citation of ensemble learning is only 38 and in year 2024 the citation increased significantly to 979 and its R^2 value i.e., 0.9292 clearly indicates an upward trend. In Fig. 4 illustrates the citation trends pertaining to ensemble learning-based disease prediction. It demonstrates a consistent rise in citations of heart disease, diabetes, liver disease, renal disease, and skin cancer from 2020 to 2024. Table 1 lists the different types of publications every year. Data shows that articles

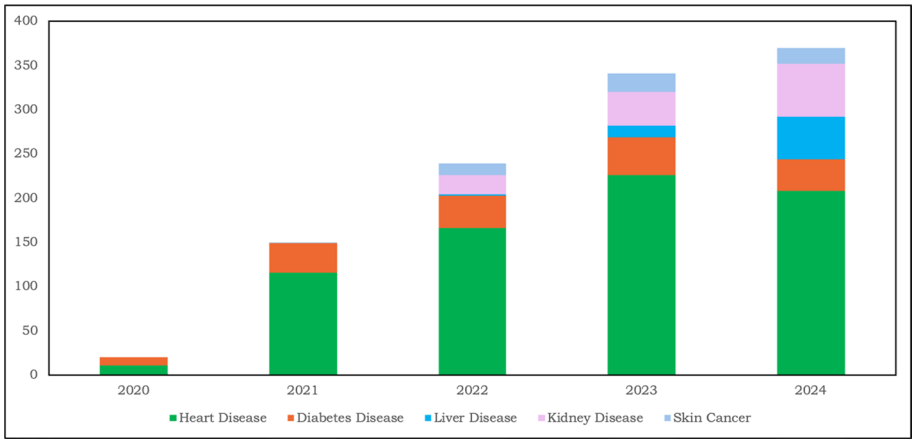


Fig. 4 Citation trends for diverse diseases utilizing ensemble learning from 2020 to 2024

Table 1 Document type distribution of publications 2004 to 2024

Document type	No. of publications	Percentage (%)
Article	24,535	60.83
Conference paper	13,081	32.43
Conference review	1067	2.64
Book chapter	764	1.90
Review	639	1.58
Book	106	0.26
Retracted	60	0.14
Letter	50	0.12
Editorial	16	0.03
Short survey	14	0.03
Total	**40,332**	**100.00**

and conference papers constitute the majority of contributions. Figure 5 represents the global position of ensemble learning research. China occupies the foremost position in this field, with roughly 8,878 papers, while India ranks second with around 8,300 publications, demonstrating their significant contributions to this discipline. Table 2 represents the top 10 application area of ensemble learning till the end of 2024. It can be noted that ensemble techniques are mostly considered for healthcare applications followed by Finance applications.

These statistical data clearly indicate the rapid growth of research in ensemble learning, driven by their capacity to improve predictive performance. In 2024, the expected count of articles utilizing "Ensemble Learning", "Ensemble Deep Learning" and "Ensemble Machine Learning" was 5,069 and 1,230 documents, respectively. Moreover, these approaches have been extensively implemented across several domains, with computer science in the forefront, exhibiting the greatest utilization rates of 33.5% for ensemble learning and 35.1% for ensemble machine learning.The growing availability of large-scale biomedical data and established effectiveness of ensemble methods underscore an imperative need to compile and analyze the existing developments in the field. So far, significant results have been reported by multiple studies, and yet there isn't a proper review that describes the state-of-the-art disease prediction ensemble methods. This motivated us to conduct an extensive review to provide a unified and up-to-date

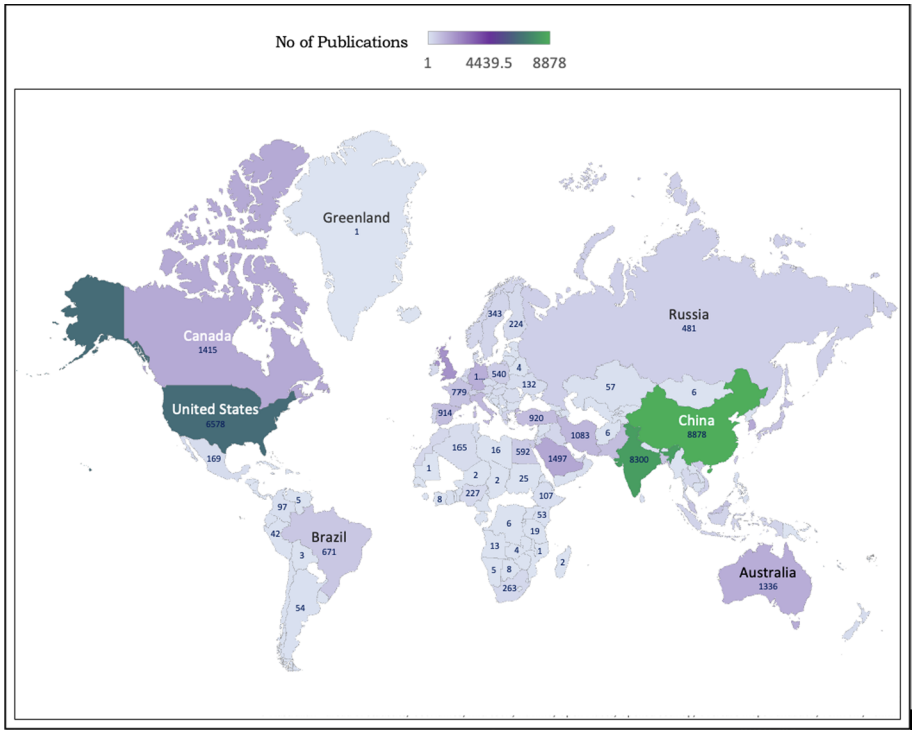


Fig. 5 World-wide publication of ensemble learning from 2004 to 2024

Table 2 Ensemble learning applications and their importance

Applications	Importance (%)
Healthcare	90
Finance	85
Marketing & retail	80
Image & video processing	95
Natural language processing	75
Cybersecurity	88
Climate & environmental studies	83
Autonomous systems	92
Scientific research	86
Education & recommendation systems	79

perspective on the role of ensemble learning in disease prediction, focusing on its applications, advantages, challenges, and future directions.

2.1 Challenges of ensemble learning

In the medical field, ensemble learning is an innovative and powerful method that integrates the strengths of multiple ML models to develop predictions that are more dependable and precise. In doing so, ensemble methods, including bagging, boosting, and stacking, are able to take advantage of diverse models like DTs, SVMs, and NNs to manage complex and imbalanced healthcare datasets. These techniques have proven very successful to predict various kinds of diseases ranging from diabetes to cardiovascular diseases and cancers, up to neurological disorders. For example, boosting techniques including AdaBoost, XGBoost have proven a good prediction result for the classification of breast cancer diagnosis and even predicting the patients' survival

percentage. Likewise, random forests are commonly utilized for the early diagnosis of ailments such as diabetes and heart conditions. With robustness improvement and minimization of overfitting, ensemble learning enables healthcare professionals with decision support through early intervention as well as personally tailored treatment programs.

Despite advantages, ensemble learning faces several hurdles that can critically reduce its effectiveness in application and implementations shown in Fig. 6:

Computational Complexity: In Bagging, boosting, or stacked ensemble methods the integration of multi-models proves to be high computationally extensive. This hampers efficient training and making predictions, usually too slow even for real time applications or highly larger datasets without superior hardware.

Risks of Overfitting: Ensemble approaches typically mitigate the risk of overfitting. Improper hyperparameter tuning, including the selection of the number of base models and learning rates in boosting, can substantially result in overfitting for noisy or inadequately large datasets.

Interpretability and transparency : Ensemble models, particularly intricate variants such as RF or GB, are predominantly regarded as black-box systems. This deficiency in interpretability may diminish their acceptance across various domains, especially in critical applications like healthcare or finance, where comprehending model decisions is essential for trust and regulatory adherence.

Data Imbalance and Quality: Ensemble methods are badly suited to imbalanced data with a particular dominant class. Poor quality such as missing or noisy values in the data would affect all the base models negatively and reduce the overall effectiveness of the ensemble. Model Selection and Diversity: The variety of fundamental models determines the success of ensemble learning. Choosing appropriate models and ascertaining that their outputs are not overly correlated poses a challenge. Lack of diversity may only lead to minor improvements in performance.



Fig. 6 Key challenges of ensemble learning

Resource and Implementation Costs : It requires high expertise, time, and financial investment to deploy and maintain ensemble models in real-world applications. The complexity of integrating multiple models into production pipelines can further complicate implementation.

Scalability issues : When the dataset or the number of base models grows, scalability problems occur. Ensemble methods may not be as useful in large systems due to memory and parallelisation issues.

Addressing these challenges would thus require careful design, optimization, and domain expertise to achieve the full potential of ensemble learning without compromising efficiency.

3 Methodology

This study provided a comprehensive review of the scientific literature to analyze the predictive modeling algorithms used for five major chronic diseases. Research publications published from 2020 to end of the 2024 were analyzed from reputable databases, including IEEE, ScienceDirect, PubMed, Springer, and Scopus. Following the significant improvements in Ensembel learning illustrated in Fig. 3, we applied the PRISMA framework [49], represented in Fig. 8, to select a targeted collection of ensamble used in prior disease prediction of esteemed works for comprehensive examination. The proposed technique is divided into three main steps, as outlined below.

3.1 Identification phase

This section identifies appropriate keywords and their synonyms as focused search terms for locating and extracting documents from several databases. The keywords were finally used for searches in the Scopus databse, Google Scholar, science direct, IEEE and others renowned database, with the search time limited from 2020 until the end of 2024. TThe search query was executed utilizing the subsequent keywords: "Disease prediction" AND "ensemble machine learning," "Disease prediction" AND "ensemble method," "Disease diagnosis" AND "ensemble learning," "Disease diagnosis" AND "ensemble machine learning," "Disease diagnosis" AND "ensemble method," "Disease diagnosis" AND "ensemble machine learning," "Disease diagnosis" AND "ensemble machine learning method," "Disease diagnosis" AND "ensemble machine learning method"," Disease prediction" AND " ensamble deep network", " Disease prediction " AND " deep learning" and " Disease diagnosis " AND " ensemble deep method". This search process led to the identification of a total of 257 relevant documents. The majority were journal articles, accounting for 106 papers (45%), followed by conference papers, which comprised 95 papers (45%). conference review papers represented 25 papers (11%), while book chapters formed a smaller portion with 6 papers (2%) and review article is only 4. as illustrated in the pie chart in Fig. 7.

3.2 Evaluation phase

This process had three screening stages to assess the found papers and verify their eligibility. The Rayyan tool [50] was employed to assist with citation sharing and to allow for the comparison of inclusion or exclusion determinations. During the preliminary evaluation, duplicate manuscripts and those originating from non-academic sources were removed. Seven duplicate manuscripts were specifically detected from the Scopus, IEEE,

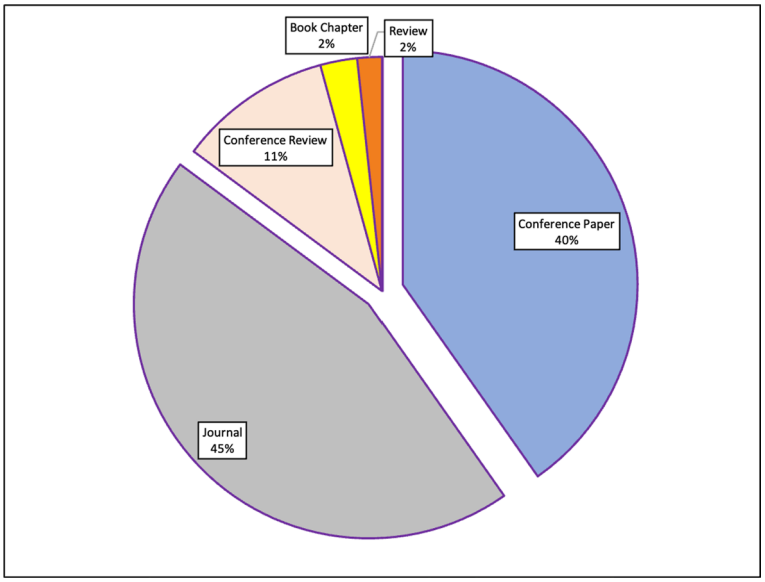


Fig. 7 Varity of pubished document from 2020 to 2024

Springer, Web of science and other reputed databases and subsequently excluded. The remaining 252 publications were subsequently examined to exclude those disseminated in non-academic or substandard journals. Figure 8, highlight that 204 articles from diverse esteemed journals were chosen for the subsequent screening phase.

During the second screening phase, the titles and abstracts of the selected articles were thoroughly examined, resulting in the elimination of 154 papers considered irrelevant to the study. This procedure resulted in 132 documents for additional evaluation. Two phases formed the final screening process: first, the full texts of the surviving papers were carefully reviewed to determine whether they met the criteria for acceptance. The following are the criteria for being part of this study:

C_1: Paper publisehed with in 2020 to end of 2024.

C_2: Paper writen in English language.

C_3: Short communication papers, and book chapters are excluded.

C_4: Published in reputed journal.

C_5: Published for disease prediction (incude like heart, diabetes, liver, kidney, and skin) using ensemble machine learning.

At this stage, the previously specified precise needs were meticulously reassessed by an in-depth review of the entire texts. Consequently, 87 papers were considered eligible. Furthermore, when examining the references listed in these articles during the second phase of the third screening, four additional papers that fulfilled the inclusion requirements were located, increasing the total to 91.

3.3 Reporting phase

In the reporting phase, a total collection of 91 pertinent publications obtained from Rayyan was classified according to six principal publishers and additional sources to meet the eligibility criteria. The distribution of eligible papers among the publishers is as follows: Elsevier (29), IEEE (14), MDPI (17), Springer (16), Wiley (8), Taylor & Francis (1), and other sources (3). The research findings were further analysed and systematically categorised into four primary groups: Heart disease predictiong using Ensemble

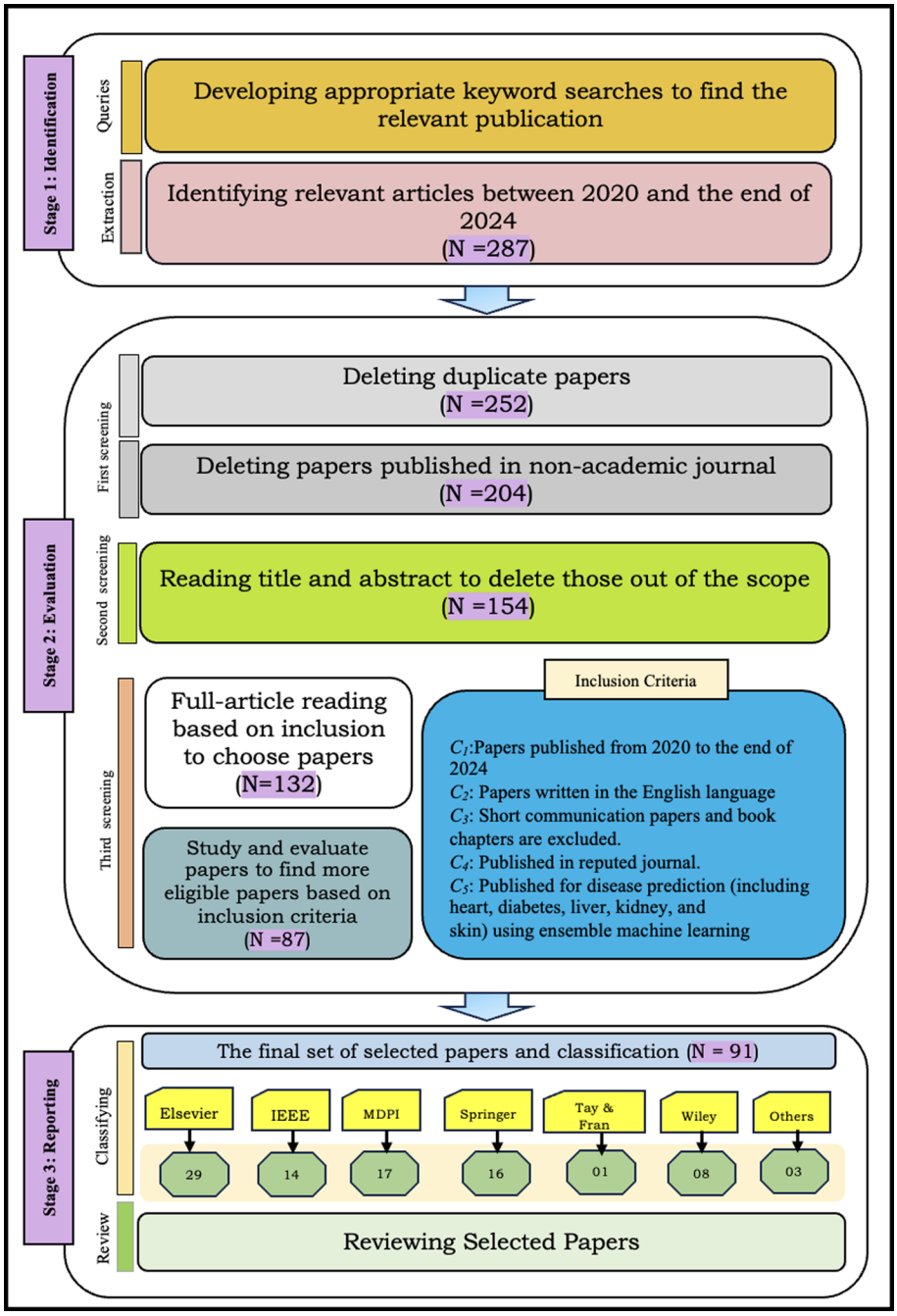


Fig. 8 Methodology of systematic review

learning, Diabetes disease predictiong using Ensemble learning, Liver disease predictiong using Ensemble learning, Kidney disease predictiong using Ensemble learning and Skin disease predictiong using Ensemble learning.

The findings were documented in the MS Office documents and utilized as reference sources for the responses addressed in the next part. This systematic review of the literature study presents a series of Research Questions (RQ) aimed at enhancing comprehension of the subject and directing future research paths. The recommended RQs are as follows:

RQ1: How have ensemble learning techniques been applied between 2020 and 2024 for predicting major diseases such as heart, diabetes, liver, kidney, and skin diseases? This question is addressed through the year-wise and disease-wise review of 91 eligible studies summarized in Sect. 3, highlighting application trends and methodological diversity.

RQ2. Which ensemble learning approaches (machine learning, deep learning, and hybrid ensembles) demonstrate superior predictive performance across different disease domains? Sect. 3 and its subsections analyze and compare ensemble machine learning, ensemble deep learning, and hybrid ensemble models using performance metrics such as accuracy, precision, recall, F-measure, and processing time.

RQ3. What combinations of base learners, ensemble strategies, and feature selection techniques obtains the most robust and consistent disease prediction outcomes? This question is explored by examining stacking, bagging, boosting, and hybrid ensembles across diseases, emphasizing the role of classifier diversity, data sampling strategies, and feature selection methods in section 3.

RQ4. How do performance metrics and computational efficiency vary across diseases and ensemble methodologies? The comparative tables and framework discussed in Sect. 3 evaluate trade-offs between predictive accuracy and computational cost, particularly for real-time and large-scale healthcare applications.

RQ5. What are the common limitations, challenges, and research gaps in existing ensemble-based disease prediction studies? Sect. 5 synthesizes key issues such as data interpretability, scalability, and computational complexity, identifying persistent challenges across disease domains.

RQ6. What future research directions can enhance the applicability and reliability of ensemble learning in predictive healthcare? Sect. 6 addresses this question by proposing directions such as adaptive and hybrid ensembles, integration with federated learning and IoT-based systems, and extension to less-explored disease categories.

These research questions collectively define the problem statement and analytical focus of the review, ensuring a structured evaluation of recent advances and clearly positioning the contribution of this study within predictive healthcare research.

4 Disease-wise study of ensemble learning approaches

All the articles published between 2020 and 2024 were reviewed to predict major diseases, including heart, kidney, liver, hypertension, and diabetes-related conditions, using ensemble learning techniques. A comprehensive collection of major disease-related publications from renowned databases was classified year-wise. The total number of journals published from 2020 to the end of 2024 using ensemble learning techniques stands at 252. Among these, the year 2024 witnessed the highest number of publications addressing diseases using ensemble learning compared to other years and diseases, but as per inclusion criteria of selected eligible paper is 91 for directly trigger major disease prediction using ensemble learning as illustrated in Fig. 9.

Figure 10 illustrates how ensemble learning is integrated into a healthcare AI pipeline, beginning with heterogeneous input data from clinical repositories, electronic health records, and rule-based systems that encode medical guidelines and expert knowledge. These inputs are preprocessed and transformed through feature engineering to create clinically meaningful and diverse feature sets that support multiple learning

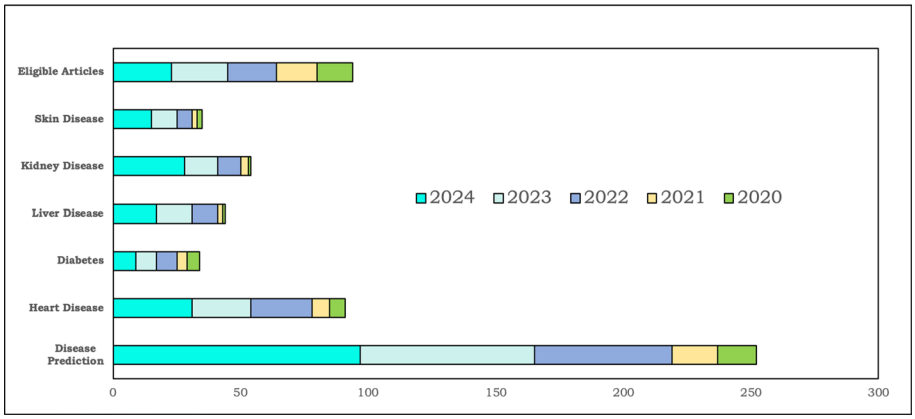


Fig. 9 Year-wise classification different major disease article and eligible articles

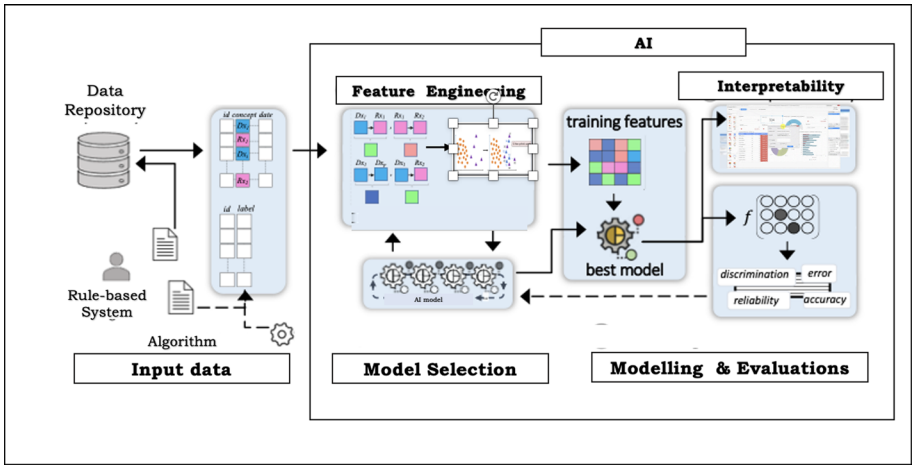


Fig. 10 A conceptual diagram for ensemble learning fits in healthcare AI

perspectives. In the model selection stage, several base models such as statistical models, tree-based methods, neural networks, and rule-based predictors are trained in parallel, and ensemble strategies like bagging, boosting, or stacking are used to combine their predictions into a more robust "best model." The modelling and evaluation phase assesses ensemble performance using healthcare-relevant metrics such as accuracy, discrimination, reliability, and error, ensuring generalizability and safety across patient populations. Finally, the interpretability layer provides transparency into ensemble decisions through explanation and visualization tools, enabling clinician trust and regulatory compliance, while feedback loops allow continuous refinement of features and models as new clinical data become available.

The Fig. 11 represented an effective framework for predicting many diseases, including heart, diabetes, liver, kidney, and skin disorders, utilising advanced computational methods to efficiently process, analyse, and interpret patient data. The process commences with the aggregation of various data types, encompassing patient information such as physiological vital human body parameters (e.g., heart rate, blood glucose levels, liver enzyme levels, renal function tests, and dermatological conditions etc.,) and health records, which comprise structured medical data including demographics, laboratory results, and medical histories. This raw data undergo an essential preprocessing phase,

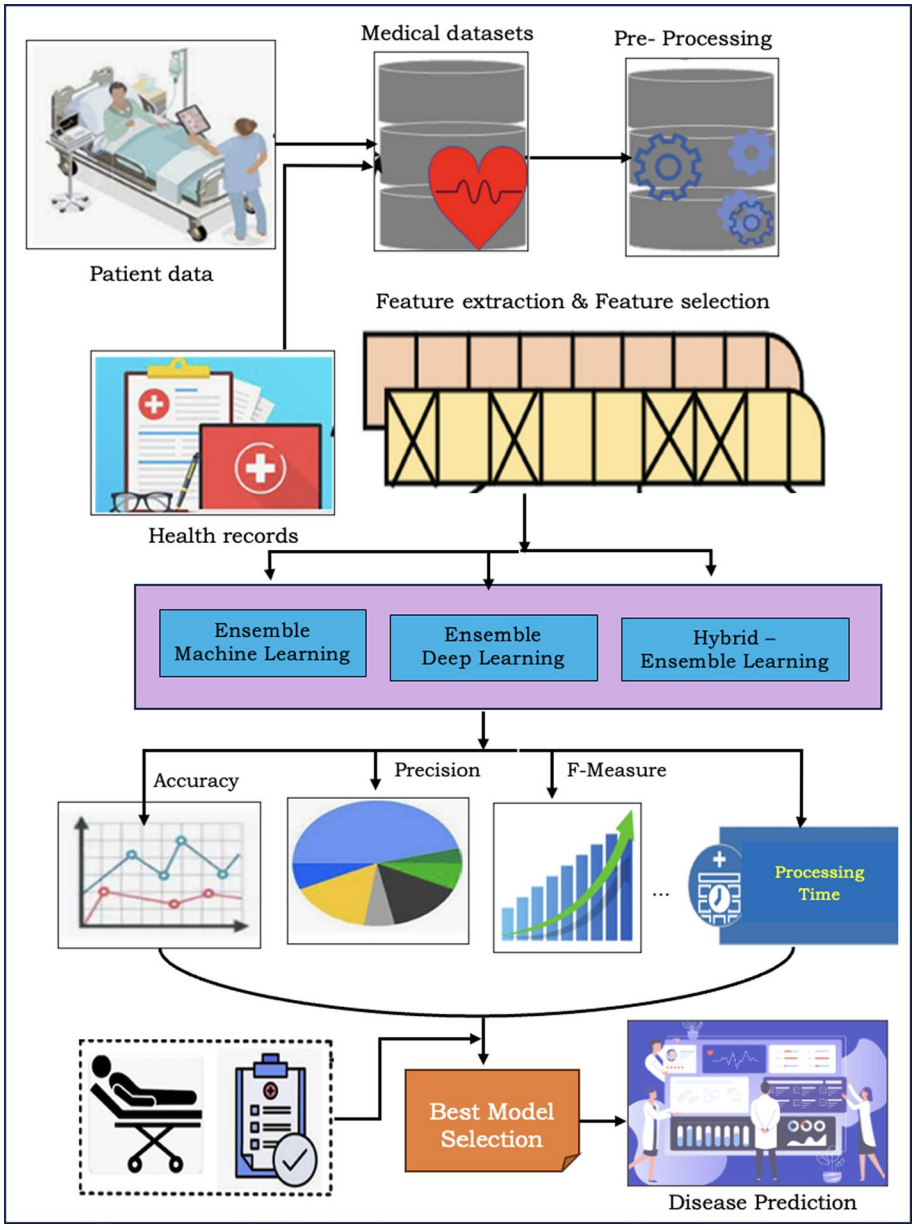


Fig. 11 A framework for disease prediction using Ensemble learning

during which significant characteristics are retrieved, and irrelevant or redundant data is discarded via feature selection techniques. This guarantees that exactly the most important factors will be used for constructing predictive models, enhancing both efficiency and precision.

The selected dataset is further examined utilising three primary computational methodologies. Ensemble deep learning techniques are employed to discern complex and non-linear patterns in large datasets, making them appropriate for difficult scenarios such as multi-disease prediction.. These methodologies generally employ neural networks capable of autonomously extracting features from unprocessed data. Conversely, ensemble machine learning techniques utilize established algorithms such as decision trees, support vector machines, and random forests, which are adept at managing structured and moderately complicated datasets. Moreover, hybrid ensemble learning

techniques are employed to improve the system's overall performance. Techniques are employed to refine hyperparameters, enhance feature selection, or augment model efficiency.

A comprehensive performance assessment is performed to guarantee the models' reliability. The analysis emphasizes essential metrics: accuracy, reflecting the model's overall correctness; precision, measuring the ratio of true positive predictions to total positive predictions; F-measure, providing a balance between precision and recall; and processing time, denoting the model's computational efficiency and suitability for real-time applications. These metrics provide a thorough evaluation of the model's performance.

The most suitable model is selected based on these assessments. This decision is essential, as it guarantees that the chosen model achieves a balance of high accuracy, precision, and computing efficiency. The chosen model is implemented for the prediction of ailments like heart disease, diabetes, liver disease, renal disease, and skin disease, facilitating precise and prompt diagnosis. Such estimates may help healthcare providers in identifying at-risk individuals and implementing preventive or remedial measures. This platform demonstrates the amalgamation of data processing, ML, and optimisation methodologies, offering a resilient, efficient, and scalable solution for tackling these significant global health issues.

The prediction of major diseases, including heart, diabetes, liver, kidney, and skin diseases, has received significant attention in recent years. The researchers have devised sophisticated computational methods and algorithms to analyse extensive information and deliver precise predictions, facilitating early diagnosis and improved treatment planning. in the next sub section review examined significant works in various fields, emphasising essential techniques and contributions of major diseases.

4.1 Heart disease prediction using ensemble learning

Heart attacks, heart failure, heart arrhythmias, congenital heart issues, and heart disease are all included in the umbrella term, cardiovascular disease. It is the leading cause of death globally and is linked to poor diet, inactivity, smoking, high blood pressure, cholesterol, obesity, and diabetes. Heart disease symptoms can include weakness, shortness of breath, chest pain, and irregular heartbeat. Maintaining a healthy lifestyle, controlling chronic diseases, frequent exercise, and avoiding tobacco reduce risk. New drugs, surgeries, and diagnostic equipment have improved heart disease care and prognosis. Out of 91 papers, 18 are reviews of heart diseases. Table 3 represented the summary of the work represented in the tabular format.

Numerous approaches for the prediction of heart disease using different datasets and ML models are provided by the evaluated papers. Mienye and Jere [51] applied an ensemble Deep Neural Network on the UCI Heart Disease dataset, achieving a remarkable 95.2% accuracy rate that was balanced across all measures but lacked feature selection. Khan et al. [52] used a Stacking approach (XGBoost, RF, SVM) on Cleveland and Statlog datasets with feature selection and achieved 94.1% accuracy with relatively good overall performance, yet recall was slightly lower by 93.5%. Nissa et al. [53] used LightGBM and AdaBoost on the UCI Heart Disease dataset and achieved accuracy to be 95%, however, missed precision, recall, and F1-score metrics. Kaur and Ahmad [54] applied XGBoost without feature selection on the same dataset, reporting a solid accuracy of 94.5%, though recall was slightly lower (93.9%) and F1-score was absent. Asif et al. [55]

Table 3 Comparative summary of heart disease prediction studies

Author	Year	Method	Dataset	Feature Selections	Acc.	Prec.	Rec.	F1	Advantage	Disadvantage
Mienye and Jere [51]	2024	Deep Neural Network Ensemble	UCI Heart Disease	N	95.2%	95.8%	94.6%	95.2%	High accuracy, robust predictions	Low interpretability, high computation
Khan et al. [52]	2024	Stacking (XGBoost, RF, SVM)	Cleveland, Statlog	Y	94.1%	94.7%	93.5%	94.1%	Improved generalization	Complex model design
Nissa et al. [53]	2024	LightGBM and AdaBoost	UCI Heart Disease	N	95.0%	–	–	–	Fast training, scalable	Limited interpretability
Kaur and Ahmad [54]	2024	XGBoost	UCI Heart Disease	N	94.5%	95.1%	93.9%	–	Strong performance on tabular data	Sensitive to hyperparameters
Asif et al. [55]	2023	CNN–LSTM Ensemble	Cleveland, Statlog	Y	96.3%	96.8%	–	–	Captures spatial and temporal patterns	Requires large datasets
Kumar et al. [56]	2023	Hybrid Stacking (RF, SVM, LR)	Cleveland, Statlog	Y	94.5%	95.1%	94.0%	94.5%	Balanced accuracy and stability	Increased computational cost
Sheikh et al. [57]	2023	Bagging (Decision Trees)	UCI Heart Disease	N	88.5%	89.2%	87.8%	88.5%	Simple, reduces variance	Lower accuracy
Samuwiza et al. [58]	2023	RF–SVM Ensemble with FS	Cleveland	Y	95.8%	96.2%	95.4%	95.8%	Effective feature reduction	Feature selection dependency
Wei et al. [59]	2023	Stacking (XGBoost, LightGBM, CatBoost)	UCI Heart Disease	N	94.8%	–	–	–	Strong ensemble diversity	High tuning complexity
Almulihi et al. [60]	2022	Deep Learning Ensemble	Statlog	N	93.9%	94.4%	93.4%	–	Robust deep feature learning	Poor explainability
Mahesh et al. [61]	2022	AdaBoost with SVM	Cleveland	Y	89.2%	89.7%	–	–	Boosts weak learners	Sensitive to noise

Table 3 (continued)

Author	Year	Method	Dataset	Feature Selections	Acc.	Prec.	Rec.	F1	Advantage	Disadvantage
Liu et al. [62]	2022	Bagging with RF	UCI Heart Disease	N	91.8%	–	–	–	Improved stability	Moderate performance
Valarmathi and Sheela [63]	2021	XGBoost Boosting	Cleveland	Y	93.7%	94.2%	–	–	Efficient gradient boosting	Hyperparameter sensitivity
Gao et al. [64]	2021	DBN–SVM Ensemble	UCI Heart Disease	Y	96.08%	97.10%	–	–	High discriminative power	Complex architecture
Mienye and Sun [65]	2021	PSO-based Stacked Autoencoder	Framingham, Cleveland	Y	97.30%	96.10%	–	–	Excellent optimization capability	High computational cost
Wang et al. [66]	2020	Stacking with LR	Z-Alizadeh Sani CHD	Y	95.43%	–	95.84%	94.44%	Simple meta-classifier	Limited nonlinearity
Mienye et al. [67]	2020	Weighted Aging Ensemble	Cleveland, Framingham	N	93.0%	91.0%	–	–	Adaptive weighting strategy	Dataset-dependent performance
Ali et al. [68]	2020	Ensemble Deep Learning	UCI Heart Disease	Y	95.8%	–	–	–	Strong representation learning	Limited interpretability

integrated CNN and LSTM with feature selection on Cleveland and Statlog datasets that achieved the highest accuracy with 96.3% and precision with 96.8%, but the recall and F1-score were not provided. Kumar et al. [56] implemented Hybrid Stacking (RF, SVM, LR) on the same datasets with feature selection which improved an 94.5% accuracy along with a well-balanced 94.5% F1-score. Sheik et al.[57]applied Bagging with Decision Trees on the UCI Heart Disease dataset without feature selection, and this has brought the lowest accuracy to be 88.5%, which indicates that this approach is limited. Samuwiza et al. [58] applied the combination of RF and SVM together with feature selection on the Cleveland dataset and resulted in 95.8% accuracy with robust performance for all metrics. Wei et al. [59] applied Stacking (XGBoost, LightGBM, CatBoost) on the UCI Heart Disease dataset without feature selection, achieving 94.8% accuracy but lacking most metrics. Almulihi et al. [60] implemented Ensemble of Deep Learning Models on the Statlog dataset. Though they achieved 93.9% accuracy, the recall and F1-score were not reported. T. R. Mahesh et al. [61] have used AdaBoost with SVM in the Cleveland dataset that contains feature selection but gained an accuracy of only 89.2%, meaning it may not be best suitable for this kind of problems. Liu et al. [62] used Bagging with RF on the UCI Heart Disease dataset, applying no feature selection, to report 91.8% accuracy but no other measures. Valarmathi and Sheela [63] used XGBoost with feature selection

on the Cleveland dataset to get 93.7% accuracy but did not state precision and recall values completely. Gao et al. [64] incorporated a Deep Belief Network with SVM, using feature selection on the UCI Heart Disease dataset, which yielded a high accuracy of 96.08% and high precision at 97.1%.Mienye and Sun [65] introduced PSO-based Stacked Sparse Autoencoder on Framingham and Cleveland datasets using feature selection, which showed the highest overall accuracy as 97.3%, though recall and F1-score metrics were missing. Wang et al. [66] used Stacking with LR on the Z-Alizadeh Sani CHD dataset using feature selection and obtained 95.43% accuracy, though precision and F1-score were not reported. Mienye et al. [67] used the Weighted Aging Classifier Ensemble on the Cleveland and Framingham datasets without feature selection, and without precision or recall, achieved 93% accuracy. Ali et al. [68] implemented feature selection to apply an ensemble of deep learning models with a 95.8% accuracy rate on the UCI Heart Disease dataset.Feature selection improved the performance in most cases, but computational overhead and missing metrics are common problems.

4.2 Diabetes disease prediction

The body's inability to produce or use insulin causes diabetes, which raises blood sugar levels. There are three primary forms of it: the autoimmune condition known as Type 1 diabetes attacks cells that produce insulin; Type 2 diabetes, caused by obesity and inactivity; and gestational diabetes, which occurs during pregnancy. Increased thirst, frequent urination, exhaustion, and weight loss are common complaints. Diabetes can cause heart disease, kidney failure, nerve damage, and eye difficulties. Effective management includes healthy nutrition, frequent exercise, blood sugar monitoring, and medicines or insulin therapy. Prevention and delay of consequences require early diagnosis and lifestyle changes. In this section, reviewed 17 article from eligible papers. The summerise of the diabets prediction represented in Table 4.

We reviewed works which comprises diverse ensemble learning techniques. To predict Type II diabetes authors considered benchmark datasets such as Western Medicine Diabetes, Pima Indians Diabetes, and Chinese datasets specific to hospitals. Oliullah et al. [69], 92.91% accuracy was achieved using stacking without feature selection on the Pima Indians Diabetes dataset; however, precision, recall, and F1-score are not reported. Sampath et al. [70] applied an ensemble of AdaBoost and XGBoost with feature selection on the same dataset with 90.04% accuracy and recall (91.10%) but did not obtain F1-score. Jain et al. [71] utilized Soft Voting that includes feature selection with a dataset of Dr. Reddy's Lab, achieving accuracy of 86.40% along with balanced metrics that included an F1-score of 86.0%.Kumar et al. [72] used a deep ensemble with feature selection on Pima Indians Diabetes, obtaining 95.26% accuracy, high recall (96.81%), and excellent F1-score (97.72%). Zhang et al. [73] applied Stacking ensemble on Pima Indians Diabetes and Chinese and Western Medicine datasets with feature selection and obtained 93.09% accuracy with well-balanced performance across metrics. Qi et al. [74] had employed Soft Voting ensemble on Pima Indians Diabetes with feature selection with a partial accuracy at 93.50%, with other metrics with notable F1-score at 98.0%. Zhou et al. [75] employed Stacking on the same dataset with DT, kNN, NB, SVM using feature selection achieving accuracy of 98.0% and showed to have strong performance over all the metrics. Dogru et al. [76] also obtained similar accuracy of 98.0% by using Stacking model on a US hospital dataset with feature selection, although the metrics were not elaborated in

Table 4 Comparative summary of diabetes disease prediction using ensemble learning approaches

Author	Year	Method	Dataset	Feature Selections	Acc.	Prec.	Recall	F1	Advantages	Disadvantages
Oliullah et al. [69]	2024	Stacking (XGBoost, NGBoost, Bagging, LightGBM, AdaBoost)	Pima Indians Diabetes	N	92.91%	–	–	–	High predictive performance; combines diverse boosting models	Increased complexity and reduced interpretability
Sampath et al. [70]	2024	Ensemble of AdaBoost and XGBoost	Pima Indians Diabetes	Y	90.04%	90.02%	91.10%	–	Robust to overfitting; improved generalization	Requires careful hyperparameter tuning
Jain et al. [71]	2024	Soft Voting (RF, Extra Trees, MLP)	Dr. Reddys' Lab Dataset	Y	86.40%	85.40%	86.40%	86.0%	Simple ensemble strategy; improved stability	Performance depends on base model diversity
Kumar et al. [72]	2024	Deep Ensemble (RF, DT, NN, AdaBoost, SVM, XGBoost)	Pima Indians Diabetes	Y	95.26%	90.72%	96.81%	97.72%	Excellent accuracy; captures complex patterns	High computational cost and limited explainability
Zhang et al. [73]	2024	Stacking (AdaBoost, SVM, MLP, kNN, RF, LR)	Pima, Chinese & Western Medicine Diabetes	Y	93.09%	93.22%	91.60%	92.25%	Strong generalization across datasets	Complex training pipeline
Qi et al. [74]	2023	Soft Voting (SVM, RF, kNN)	Pima Indians Diabetes	Y	93.50%	–	85.0%	98.0%	Balanced performance; low variance	Limited performance gain over strong single models

Table 4 (continued)

Author	Year	Method	Dataset	Feature Selections	Acc.	Prec.	Recall	F1	Advantages	Disadvantages
Zhou et al. [75]	2023	Stacking (DT, kNN, NB, SVM)	Pima Indians Diabetes	Y	98.0%	96.2%	98.2%	97.1%	Very high accuracy; effective feature integration	Risk of overfitting on small datasets
Dogru et al. [76]	2023	Stacking (LR, DT, RF, GB, SVM)	130 US Hospital Dataset	Y	98.0%	–	–	–	Strong clinical performance on large dataset	Reduced transparency for clinical adoption
Abnoosian et al. [77]	2023	Weighted Ensemble (kNN, SVM, DT, RF, AdaBoost, GNB)	Iraqi Diabetic Patients	Y	98.87%	98.61%	97.92%	98.51%	High reliability and balanced metrics	Dataset-specific generalization concerns
Dutta et al. [78]	2022	Weighted Ensemble (DT, RF, XGB, LGB)	BDDCD	Y	73.5%	–	–	–	Handles imbalanced data effectively	Lower accuracy compared to other studies
Abdollahi et al. [79]	2022	Hybrid Stacking (SVM, kNN, MLP, DT, ADB, NB, GA)	Diabetes 130-US Hospitals	Y	99.06%	–	99.0%	100.0%	Exceptional predictiveperformance	High model and computational complexity
Mahesh et al. [80]	2022	Blended Ensemble (LR, DT, SVM, kNN, RF)	Unspecified	N	97.11%	96.96%	97.96%	97.46%	Balanced accuracy and recall	Dataset details not clearly reported
Rajendra and Latifi [81]	2021	Max Voting (LR, DT, SVM, kNN, NB)	Pima Indians Diabetes	Y	88.89%	–	–	–	Simple and computationally efficient	Limited improvement over individual classifiers

Table 4 (continued)

Author	Year	Method	Dataset	Feature Selections	Acc.	Prec.	Recall	F1	Advantages	Disadvantages
Kumari et al. [82]	2021	Soft Voting (LR, NB, RF)	Pima Indians Diabetes	N	79.08%	73.13%	70.0%	71.56%	Easy implementation	Relatively low predictive performance
Hasan et al. [83]	2020	Weighted Soft Voting (kNN, DT, RF, NB, MLP)	Pima Indians Diabetes	Y	85.9%	–	78.90%	93.4%	Improved recall and robustness	Moderate accuracy
Singh and Singh [84]	2020	Stacking-based Evolutionary Ensemble (kNN, SVM, NN, DT)	PID	N	93.8%	96.1%	79.9%	88.5%	Optimized ensemble selection	Increased training time
Sarwar et al. [85]	2020	Hybrid Ensemble (ANN, SVM, kNN, NB)	Pima Indians Diabetes	N	98.60%	–	–	–	High diagnostic accuracy	Limited explainability

detail. Abnoosian et al. [77] achieved the highest of 98.87% accuracy applying weighted ensemble with feature selection on Iraqi diabetic patients, which had balanced metrics. Dutta et al. [78] performed only 73.5% with weighted ensemble on BDDCD dataset and reported the entire process as being badly dataset-dependent. Abdollahi et al. [79] presented the best accuracy of 99.06% with Hybrid on a US hospital dataset while applying feature selection and achieved perfect 99% recall and 100% F1-score. Mahesh et al. [80] performed a Blended Ensemble on an unspecified dataset that did not apply feature selection and achieved 97.11% accuracy. Rajendra and Latifi [81] applied Max Voting and feature selection on Pima Indians Diabetes, achieving 88.89% accuracy but did not provide detailed metrics. Kumari et al. [82] used Soft Voting without feature selection and achieved only 79.08% accuracy. Hasan et al. [83] applied Weighted Soft Voting on Pima Indians Diabetes with feature selection and achieved 85.9% accuracy with F1-score (93.4%) that shows class imbalance handling. Singh and Singh [84] were able to obtain 93.8% accuracy through Stacking-based Evolutionary Ensemble without feature selection that report very high precision at 96.1% but poor recall at 79.9%. Lastly, Sarwar et al. [85] used Hybrid Ensemble without feature selection, obtaining 98.60% accuracy while missing other metrics. To sum up, feature selection improved performance across all but still has some metric gaps and dataset-specific challenges.

4.3 Liver disease prediction

Liver diseases are problems that affect the liver's vital functions, such as detoxification, protein synthesis, and the formation of biochemicals needed for digesting. These diseases include hepatitis caused by viruses or toxins, cirrhosis caused by chronic liver damage and resulting in scarring and liver failure, hepatocellular carcinoma (HCC), the most prevalent form of liver cancer, and non-alcoholic fatty liver disease (NAFLD). Early detection and accurate diagnosis are a couple of factors that improve efficient management and treatment. Recently, ensemble learning techniques are getting popularity in predicting and classifying liver diseases by using various datasets and methodologies. For the review purpose, we selected 18 reputed papers from 2020 to the end of 2024. For more understandable information, the summarized details of liver diseases, including key charactersistics are presented in Table 5.

Ahad et al. [86] applied a feature-selection stacking ensemble on the Hepatitis C dataset, achieving the best recorded accuracy of 99.84%; however, F1-score, precision, and recall are not given. Ganie and Pramanik [87] used boosting approach by combining multiple boosting algorithms without using any feature selection on the LDPD and achieved an 98.80% accuracy with balanced metrics at precision and 99.0% recall values and 97.74% F1-score, Guo et al. [88]used stacking approach on Steatotic Liver Disease Image dataset with an accuracy of 91.34%. Badvath et al. [89] improved an accuracy of 99.18% with ensemble LR and NB using feature selection on Cirrhosis Liver Disease, Alizargar et al. [90] has applied a comprehensive ensemble on NHANES dataset with feature selection in order to achieve balanced performance across all metrics (99%). Theerthagiri [91] applied Histogram-based Gradient Boosting ensemble with feature selection on Indian Liver Patient Records dataset, achieving an accuracy of 94.0%. The recall value is just slightly lower, at 91.29%. Md Abdul Quadir et al. [92] used stacking with feature selection on the Indian Liver Patient Dataset (ILPD), achieving a balanced performance with an 91.82% accuracy and an 91.75% F1-score. Nakata and Siina [93] used voting ensembles on ultrasound images, with an accuracy of 71.9% and precision at 73.1%.Anisha et al. [94] presented high performance with a heterogeneous ensemble of DenseNet and ResNet on CT images with feature selection, resulting in balanced metrics. Dritsas and Trigka [95] achieved 80.1% accuracy with an ensemble voting method on ILPD without feature selection. Alotaibi et al. [96] considered bagging on the UCI HCV dataset with feature selection, achieving 96.92% accuracy with high precision. Zhao and Pan [97] used bagging on the BUPA Liver Disorder dataset with a low accuracy value of 80.3%, with a lag for precision and recall values, reflecting problems with the dataset. Meng et al. [98] attained 99.1% accuracy in UK Biobank data set using stacking. Again, recall and F1-score were not reported. Sharma and Kumar [99] applied gradient boosting to HCC with a recall of 94.73% and an accuracy of 93.92%. Chicco and Jurman [100] used an RF-based ensemble with feature selection on Hannover Medical School's dataset, where they achieved 95.4% accuracy and high precision. Singh et al. [101] used bagging and boosting (DT, J48) on the Hepatitis dataset without feature selection, achieving accuracy of 99.35%, and got balanced F1-scores. Padmakala et al. [102] achieved 94.5% accuracy with an ensemble SVM on the UCI HCV dataset, reporting recall of 92.675%. ksiazet et al. [103] used stacking on CHUC's HCC dataset with feature selection, achieving 90.3% accuracy and an 88.57% F1-score.

Table 5 Comparative summary of liver disease prediction using ensemble learning approaches

Author	Year	Method	Dataset	Feature Selections	Acc.	Prec.	Recall	F1	Advantages	Disadvantages
Ahad et al. [86]	2024	Stacking (LR, SVM, DT, RF, XGBoost)	Hepatitis C dataset	Y	99.84%	–	–	–	Extremely high accuracy; effective model fusion	Complex architecture; limited interpretability
Ganie and Pramanik [87]	2024	Boosting (LogitBoost, GB, AdaBoost, XGBoost, CatBoost, LGBM)	LDPD	N	98.80%	99.0%	99.0%	97.74%	Strong generalization and robustness	Computationally expensive
Guo et al. [88]	2024	Stacking (VGG16, RF, SVM, LR)	Steatotic liver images	N	91.34%	–	–	–	Combines deep and classical models effectively	High training cost; limited transparency
Badvath et al. [89]	2024	Ensemble (NB, LR)	Cirrhosis dataset	Y	99.18%	–	99.12%	98.92%	Simple ensemble; high predictive accuracy	Limited model diversity
Alizargar et al. [90]	2024	Ensemble (RF, kNN, LR, SVM, XGBoost)	NHANES	Y	99.0%	99.0%	99.0%	99.0%	Excellent performance across all metrics	Potential overfitting risk
Theerthagiri [91]	2024	Histogram-based Gradient Boosting	ILPD	Y	94.0%	93.50%	91.29%	–	Efficient handling of structured data	Moderate recall compared to accuracy
Md et al. [92]	2023	Stacking (XGBoost, RF, Extra Trees)	ILPD	Y	91.82%	92.72%	91.89%	91.75%	Balanced performance and robustness	Increased training complexity

Table 5 (continued)

Author	Year	Method	Dataset	Feature Selections	Acc.	Prec.	Recall	F1	Advantages	Disadvantages
Nakata and Siina [93]	2023	Voting Ensemble	Ultrasound images (USA)	N	71.9%	73.1%	71.9%	–	Simple ensemble design	Relatively low predictive accuracy
Anisha et al. [94]	2023	Heterogeneous DL Ensemble (DenseNet, ResNet)	CT images	Y	98.6%	98.1%	98.22%	98.7%	Strong feature representation highlighting	High computational demand
Dritsas and Trigka [95]	2023	Voting Ensemble (NB, SVM, LR, ANN)	ILPD	N	80.1%	80.4%	80.1%	80.1%	Improved stability over single models	Limited performance gain
Alotaibi et al. [96]	2023	Bagging (RF, XGBoost, Extra Trees)	UCI HCV	Y	96.92%	99.81%	94.0%	–	High precision and explainability	Lower recall imbalance
Zhao and Pan [97]	2022	Bagging (XGBoost, SVM, RF)	BUPA	N	80.3%	68.75%	38.82%	–	Handles variance effectively	Very low recall
Meng et al. [98]	2022	Stacking (RF, ANN, MLP, LR)	UK Biobank	N	99.1%	91.2%	–	–	Strong population-level prediction	Limited metric reporting
Sharma and Kumar [99]	2022	Gradient Boosting Ensemble	HCC dataset	N	93.92%	–	94.73%	93.0%	Effective for nonlinear patterns	Interpretability challenges
Chicco and Jurman [100]	2021	RF-based Ensemble	Hannover Medical School	Y	95.4%	97.0%	–	–	Strong feature importance analysis	RF bias toward dominant features
Singh et al. [101]	2021	Bagging & Boosting (DT, J48)	Hepatitis dataset	N	99.35%	99.4%	–	99.4%	Very high classification accuracy	Limited scalability

Table 5 (continued)

Author	Year	Method	Dataset	Feature Selections	Acc.	Prec.	Recall	F1	Advantages	Disadvantages
Padmakala et al. [102]	2021	Ensemble SVM	UCI HCV	Y	94.5%	–	92.675%	94.12%	Improved margin-based classification	Sensitive to parameter tuning
Ksiazek et al. [103]	2020	Stacking (kNN, RF, NB, GA)	CHUC HCC	Y	90.3%	–	–	88.57%	Optimized feature–model selection	Increased computational cost

4.4 Kidney disease prediction

The loss of kidney function occurs gradually in chronic kidney disease (CKD), a condition that progresses slowly. The key to reducing its progression and enhancing patient outcomes is early detection and care. ML, especially ensemble learning techniques, has become very instrumental in predicting CKD, integrating several models to increase accuracy and robustness. In a variety of datasets, ensemble techniques including as stacking, bagging, and boosting often used in combination with feature selection have demonstrated remarkable outcomes. Here we review recent works which comprises diverse ensemble learning techniques. The Table 6 highlights multiple papers published between 2020 and 2024 that focused on the ensemble and ML techniques for different health datasets and key finding of the authors. Moreover, for kidney disease the final set of eligible paper is 19 from the reputed publications.

To begin with, Dina et al. [104] stacked CNN, LSTM, and BLSTM on NHIRD and gained 97.12% accuracy. The model demonstrated high precision (99%) and recall (98%), although the complexity would make it problematic for real-time applications. Chhabra et al. [105] have applied AdaBoost, XGBoost, and other boosting algorithms on the UCI CKD dataset and have gained 99.0% accuracy. The model balanced precision and recall at an average of 98.68% and 99.21%, respectively. Yousif et al. [106] combined LSTM, BiLSTM, and DNN with stacking, achieving a 98.11% accuracy. However, its F1-score was only 95.00%, thus showing potential imbalance in the metrics. Rahaman et al. [107] achieved a maximum 99.75% accuracy on the UCI CKD dataset with an ensemble-based LightGBM and RFE, which proved to be strong in all metrics. Venkatesan et al. [108] applied XGBoost, LR, and other techniques on the UCI CKD dataset, obtaining 95% accuracy. Even though it shows high precision of 96.30% and 97.56% recall, its 96.93% F1-score, indicating minor imbalances in the predictions. Alsekait et al. [109] applied stacking with RNN, LSTM, GRU, and SVM to obtain 99.69% accuracy, showing great precision and recall values. The computational requirements of the technique might limit its scalability. Wu et al. [110] ensemble C5.0, SVM, NB, and XGBoost to attain 81.0% accuracy on an unspecified dataset. The methodology was imbalanced as incomplete metrics were shown. Islam et al. [111] stacked using SVM, DT, and LightGBM yielded 98.3% with consistent metrics while showing robust prediction. Debal et al. [112] used XGBoost, RF, and stacking on the St. Paulo's Hospital dataset and achieved 84.78%

Table 6 Comparative summary of kidney disease prediction using ensemble learning approaches

Author	Year	Method	Dataset	Feature selections	Acc.	Prec.	Recall	F1	Advantages	Disadvantages
Dina et al. [104]	2024	Stacking (CNN, LSTM, BLSTM)	NHIRD	Y	97.12%	99%	98%	97%	Captures temporal patterns in EHR data	High computational and training cost
Chhabra et al. [105]	2024	Boosting (AdaBoost, XGB, DT, RF, LGBM)	UCI CKD	Y	99.0%	98.68%	99.21%	98.93%	Excellent performance on tabular data	Requires careful hyperparameter tuning
Yousif et al. [106]	2024	Stacking (LSTM, BiLSTM, BiGRU, KNN, DNN)	UCI CKD	Y	98.11%	98.16%	97.83%	95.00%	Strong sequential feature learning	Increased model complexity
Rahaman et al. [107]	2024	Ensemble (RFE + LightGBM)	UCI CKD	Y	99.75%	99.40%	99.61%	99.75%	High accuracy with effective feature selection	Sensitive to feature selection strategy
Venkatesan et al. [108]	2023	Ensemble (XGB, LR, DT, KNN, SVM)	UCI CKD	Y	95%	96.30%	97.56%	96.93%	Balanced precision–recall trade-off	Moderate improvement over single models
Alsekait et al. [109]	2023	Stacking (RNN, LSTM, GRU, SVM)	UCI CKD	Y	99.69%	99.71%	99.69%	98.78%	Excellent temporal modeling performance	Reduced interpretability
Wu et al. [110]	2023	Ensemble (C5.0, SVM, NB, XGB)	Unspecified	Y	81.0%	86.0%	91%	–	Improved recall through ensemble voting	Low overall accuracy
Islam et al. [111]	2023	Stacking (SVM, DT, KNN, LGBM)	UCI CKD	Y	98.3%	98.0%	98.0%	98.0%	Robust and stable performance	Increased training time

Table 6 (continued)

Author	Year	Method	Dataset	Feature selections	Acc.	Prec.	Recall	F1	Advantages	Disadvantages
Debal et al. [112]	2022	Ensemble (XGB, DT, SVM, RF)	St. Paulo's Hospital	Y	84.78%	82.89%	82.56%	–	Handles heterogeneous clinical data	Relatively lower accuracy
Silveira et al. [113]	2022	AdaBoosted Decision Tree	UFAL Hospital	Y	94.44%	95%	94.78%	90%	Lightweight and efficient boosting approach	Limited model diversity
Dritsas et al. [114]	2022	Stacking/Bagging (LR, SVM, KNN, ANN)	UCI CKD	Y	97.45%	94.67%	97%	–	Improved robustness over base learners	Complex ensemble design
Kazemi et al. [115]	2021	Ensemble (MLP, SVM, RT, Bagging, Boosting)	UCI CKD	Y	99.1%	97.34%	96.12%	96.46%	High predictive accuracy	Computationally intensive
Chaudhuri et al. [116]	2021	Enhanced Decision Tree Ensemble	UCI CKD	Y	100.0%	100.0%	100.0%	–	Perfect classification on benchmark dataset	Possible overfitting risk
Elkholy et al. [117]	2021	Deep Neural Ensemble	UCI CKD	N	98.5%	–	87.5%	–	Strong representation learning	Lower recall; limited explainability
Ventrella et al. [118]	2021	Voting (LR, DT, RF, NN)	Vimercate Hospital EMR	Y	91.0%	94.0%	94.0%	94.0%	Simple and interpretable ensemble	Moderate accuracy
Zhang et al. [119]	2021	Stacked Sparse Autoencoder	Guangdong TCM Hospital	Y	–	94.50%	96.5%	95.33%	Effective unsupervised feature learning	Accuracy not reported
Jongbo et al. [120]	2020	Bagging & Random Subspace	UCI CKD	Y	98.3%	–	100.0%	–	Excellent recall performance	Limited precision reporting

Table 6 (continued)

Author	Year	Method	Dataset	Feature selections	Acc.	Prec.	Recall	F1	Advantages	Disadvantages
Sudharson and Kokil [121]	2020	AdaBoost, Bagging (J48, KNN)	UCI CKD	Y	95.56%	93.45%	93.14%	–	Computationally efficient ensemble	Lower improvement margin
Khamparia et al. [122]	2020	Deep Stacked Autoencoder Network	UCI CKD	Y	100.0%	100.0%	100.0%	–	Excellent deep feature extraction	High training complexity

accuracy. While accurate, both precision (82.89%) and recall (82.56%) values appear to be low for generalizing. Silveria et al. [113] applied AdaBoosted decision trees on University Hospital to obtain 94.44% accuracy. The model had a very good precision of 95% but dropped to an F1 score of 90%. Dritsas et al. [114] employed stacking and bagging with LR, SVM, and KNN on the UCI CKD dataset to get 97.45% accuracy. Its effectiveness is further reflected in the precision (94.67%) and recall (97%) metrics, though there is room for metric consistency. Kazemi et al. [115] achieved 99.1% accuracy with MLP, SVM, and boosting on the UCI CKD dataset, with balanced metrics.Chaudhuri et al. [116] got excellent accuracy of 100% on the UCI CKD dataset with the ensemble-enhanced decision tree, and it was reliable. Elkholy et al. [117] have used a deep neural ensemble with 98.5% accuracy, while incomplete recall of 87.5% hampers evaluation. Ventrella et al. [118] used voting LR, DT, RF and NN together on the Vimercate Hospital EMR dataset to get 91.0% of accuracy with consistent precision and recall metrics of 94.0%.Zhang et al. [119] applied stacked sparse autoencoder on the Guangdong Provincial dataset with high recall (96.5%) but incomplete metrics. Jongbo et al. [120] used ensemble bagging on the UCI CKD dataset with a perfect recall (100%) of 98.3% accuracy. Sudharson and Kokil [121] combined AdaBoost and bagging with J48 and KNN on the UCI CKD dataset to get 95.56% accuracy with balanced metrics. Khamparia et al. [122] achieved a high score of 100% for a deep stacked autoencoder network on the UCI CKD dataset, making it highly reliable. Ensemble and the most advanced techniques for CKD prediction will continue to be hybrid approaches, with both computational efficiency as well as variance in datasets showing promise for development.

4.5 Skin cancer prediction

Skin diseases, that can be brought on by allergies, irritants, heredity, or environmental factors, include ailments including psoriasis, acne, eczema, and infections. The symptoms can include blisters or scaling, as well as rashes, redness, and itching. Effective treatments are provided by advances in dermatology, and problems can be managed and prevented with proper skincare, hygiene, avoiding triggers, and medical therapy. Early diagnosis is essential. Skin disease prediction has remarkably become more accurate and reliable through ensemble learning and advanced machine learning techniques. For reliable outcomes on datasets like HAM10000 and SIIM-ISIC, methods develop based on the application of different classifiers and optimization methodologies. Although the

techniques are highly effective, some challenges include high computational complexity and variability in performance metrics across models and datasets. In this section 19 paper is added for the eligible set for the reviewed. The key characteristic of kidney disease of the reviewed papers are discussed on Table 7.

Natha et al. [123] achieved 96.20% accuracy using a max voting ensemble on the HAM10000 dataset. Although the model showed well-balanced metrics, computational complexity is a significant issue. Alenizi et al. [124] used stacking on the SIIM-ISIC dataset with an accuracy of 87.72% and high precision (92.15%), but omitted recall and F1-score metrics, which cannot be considered a complete evaluation. Shaik et al. [125] employed stacking with CNN, ResNet50, and DenseNet for 91.73% accuracy on HAM10000 and DermNet datasets, though incompletely reported metrics blur the findings. Bhosle et al. [126] used stacking of CNN, RNN, SVM, and GANs for an accuracy of 93.68% on the HAM10000 and DermNet datasets, striking a balance between precision (91.56%) and recall (92.78%). Kalpana et al. [127] has achieved the best accuracy of 97.66% with the ensemble SVM and RF combined with hybrid Aquila optimization, having excellent recall and F1-score at 71.54% and 98.25%, respectively, though poor precision at 74.25%. Reddy et al. [128] have used ensemble E-CNN, SVM, RF, and more to achieve accuracy at 98%, but did not provide more recall metrics in detail. Yashu et al. [129] applied a hybrid deep ensemble on the SIIM-ISIC dataset, consequently, although having a somewhat lower F1-score of 94.16%, it obtained an accuracy of 98.67% and a remarkable recall of 99.94%. Liu et al. [130] used max voting and stacking on the HAM10000 dataset with an accuracy of 86.7% but presented inconsistent metrics. Ahamed et al. [131] implemented a deep CNN and stacked LSTM, which resulted in 91.73% accuracy but with incomplete metric reporting. Similarily, Menna et al. [132] used KNN, SVM and RF on 90% National Data Center Dataset with precision up to 86%. Mohhmad et al. [133] created stacked ensemble of CNN with a level of 95.76%. Precision equals recall 96.67%, however, with balanced recall equaling 95.7%. RF and a deep convolutional network Kalaivani et al. [134] had results of 96.1% on the HAM10000 dataset with all consistent precision and recall metrics. Hossain et al. [135] used weighted ensemble methods on the HAM10000 and ISIC 2019 datasets with 91% accuracy but only reported incomplete metrics. Jong et al. [136] used KDE methods on the SD-HZ dataset, which obtained 91.5% accuracy but had recall at 82%, so it is still open for improvement.Kausar et al. [137] merged majority and weighted voting with ResNet and DenseNet, attaining an 95.76% accuracy with good precision i.e., 95.60% and 96.67% recall. Verma et al. [138] attaining the accuracy of 99.67% using bagging, boosting, and stacking, which exhibits high reliability. Verma and Pal [139] achieved the accuracy of 98.35% on UCI Skin Disease dataset, focusing on high precision of 100%. Mahbod et al. [140] used multi-scale CNN with 87.42% using ISIC 2018. Tang et al. [141] used global-part CNN with ISIC 2016/2017 datasets that attained 92.6% accuracy and good recall rate of 96.5%.In general, ensemble and deep learning methods can only strengthen skin disease diagnosis, and predictive values vary depending on the use case as well as on particular datasets, feature selection, and complexities of the model.

Table 7 Comparative summary of skin cancer prediction using ensemble learning approaches

Author	Year	Method	Dataset	FS	Acc.	Prec.	Recall	F1	Advantages	Disadvantages
Natha et al. [123]	2024	Max Voting (RF, GB, AdaBoost, CatBoost, Extra Trees)	HAM10000	Y	96.20%	96.30%	95.50%	95.63%	Strong robustness through diverse tree models	Increased computational overhead
Alenizi et al. [124]	2024	Stacking (RF, XGB, MLP, SVM)	SIIM-ISIC	Y	87.72%	92.15%	–	–	Combines classical and neural models	Moderate accuracy
Shaik et al. [125]	2024	Stacking (CNN, ResNet-50, DenseNet)	HAM10000, DermNet	Y	91.73%	–	–	–	Strong feature extraction using deep CNNs	High training complexity
Bhosle et al. [126]	2024	CNN, RNN, SVM, GANs Ensemble	DermNet, HAM10000	Y	93.68%	91.56%	92.78%	92.65%	Handles complex image patterns effectively	Computationally expensive
Kalpana et al. [127]	2023	Ensemble SVM + RF with Aquila Optimization	HAM10000	Y	97.66%	74.25%	71.54%	98.25%	Optimized feature selection improves accuracy	Low recall and precision imbalance
Reddy et al. [128]	2023	E-CNN Ensemble (SVM, RF, AdaBoost, XGB, ANN)	PH2, HAM10000	Y	98%	97.38%	94.97%	–	High diagnostic accuracy across datasets	Complex ensemble pipeline
Yashu et al. [129]	2023	Hybrid Deep Ensemble	SIIM-ISIC	Y	98.67%	95.78%	99.94%	94.16%	Excellent sensitivity for melanoma detection	High computational cost
Liu et al. [130]	2023	Voting & Stacking Ensembles	HAM10000	N	86.7%	88.0%	–	86.34%	Simple ensemble improves stability	Lower predictive performance
Ahamed et al. [131]	2022	Deep CNN + Stacked LSTM	HAM10000	Y	91.73%	–	91.83%	–	Captures spatial and sequential features	High training cost
Menna et al. [132]	2022	KNN, SVM, RF Ensemble	National Data Center	Y	90%	86%	–	94%	Easy implementation and interpretability	Limited scalability
Mohhmad et al. [133]	2022	CNN-based Stacked Ensemble	Unspecified	Y	95.76%	96.67%	95.7%	95.45%	High accuracy with explainability support	Dataset details not specified
Kalaivani et al. [134]	2022	RF + Deep CNN	HAM10000	Y	96.1%	94.93%	95.17%	96.1%	Effective hybrid deep-learning approach	Reduced model transparency

Table 7 (continued)

Author	Year	Method	Dataset	FS	Acc.	Prec.	Recall	F1	Advantages	Disadvantages
Hossain et al. [135]	2021	Weighted DL Ensemble	HAM10000, ISIC 2019	N	91%	–	94%	91%	Improved recall via weighted fusion	Requires large training data
Jong et al. [136]	2021	KDE-based Ensemble (CT, AT, ST)	SD-HZ	Y	91.5%	87%	82%	84%	Robust density-based classification	Performance sensitive to kernel choice
Kausar et al. [137]	2021	Majority & Weighted Voting (ResNet, DenseNet)	HAM10000	Y	95.76%	95.60%	96.67%	96.13%	High multiclass classification accuracy	Increased computational requirements
Verma et al. [138]	2020	Bagging, Boosting, Stacking	HAM10000	Y	99.67%	95%	96.78%	96.56%	Excellent overall predictive performance	Possible overfitting on benchmark dataset
Verma and Pal [139]	2020	Stacking (NB, DT, SVM, RF)	UCI Skin Disease	Y	98.35%	100%	96%	98%	Simple yet highly effective ensemble	Dataset-specific evaluation
Mahbod et al. [140]	2020	Multi-scale Multi-CNN	ISIC 2018	N	87.42%	87.81%	88.56%	87.67%	Captures lesions at multiple scales	Lower accuracy compared to ensembles
Tang et al. [141]	2020	Global-Part CNN	ISIC 2016, 2017	Y	92.6%	–	96.5%	–	Effective global–local feature learning	Complex network architecture

5 Discussion, analysis and limitations

This systematic study examines the utilisation of ensemble learning algorithms in disease prediction, providing a comprehensive examination of their application across many healthcare sectors, such as early diagnosis, risk assessment, personalised treatment, and health monitoring. The study utilised a PRISMA technique [49] to systematically review research articles released from the inception of ensemble learning in health disease from 2020 to end of 2024. The identification, evaluation, and reporting phases made up the procedure.

In the identification step, a thorough search utilising predefined keywords uncovered research employing ensemble approaches, including bagging, boosting, stacking, random forests etc, for disease-related applications. The review step entailed stringent screening to ensure the inclusion of excellent studies specifically centred on disease prediction. The research were further classified according to particular applications, including chronic disease prediction, infectious disease detection, and genetic risk assessment. The outcomes in the reporting phase underscored the increasing significance of ensemble approaches in enhancing prediction accuracy, adaptability, and the capacity to manage complex disease data.

In Fig. 12, which illustrates the distribution of academic articles published between 2020 and 2024, a gradual increase in publications is observed between 2020 and 2021. However, from 2022 to 2024, there was a significant surge in publications, with 96 papers published in 2024 alone. This growth highlights the increasing application of ensemble

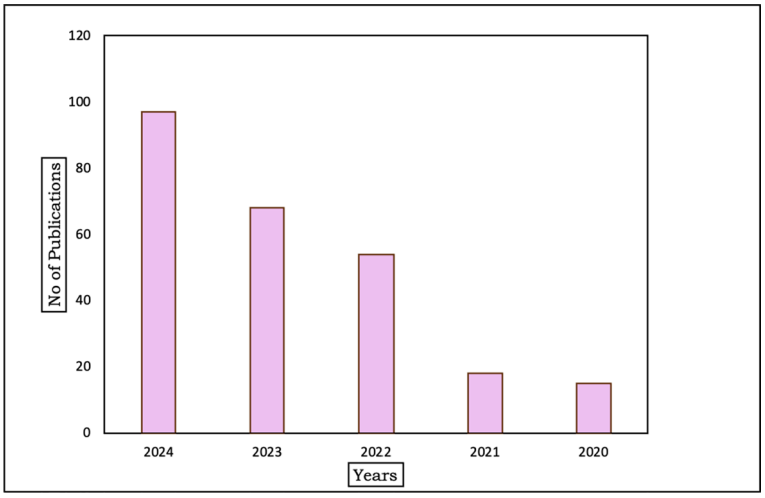


Fig. 12 Year wise publication count

learning methods in disease prediction. The reporting phase is going to summarize the main results, novel developments, and potential directions for future research in this developing field.

Subsequently, in the review phase, we implemented three screening procedures and utilised stringent inclusion criteria to identify the most pertinent and high-quality publications. We obtained these articles from reputed journals published between 2020 and the end of 2024. This procedure ensured the selection of only the most qualified and significant papers for subsequent analysis.

From the original collection of 287 documents, the evaluation step effectively selected 91 papers as the final subset. The initial screening phase eliminated ten duplicate papers, i.e., 35 and 48 publications from non-academic sources, resulting in 204 credible papers for subsequent study, and after screening the title and abstract, 52 papers were eliminated. Finally, 154 papers are selected for further evaluation to ascertain their relevance and quality in accordance with the study's aims.

Figure 13 depicts the distribution of renowned papers chosen during the initial screening, classified by respectable publishers. The publishers encompass Springer, IEEE, Elsevier, MDPI, WILEY, Taylor & Francis, and others highlighting the reputed academic sources from which the papers were generated.

Out of the 252 reputed papers, Elsevier and Springer are the highest number of publication, making up 28% and 23% of the relevant papers, respectively. This shows that important these publishers are to moving study forward in the field.

The IEEE Access publication has the most papers (25) of the 252 that were chosen for the first round of reviews. In Fig. 14, the journals with more than five publications are shown to make the most important contributions to the field.

The evaluation phase's, i.e, second stage involves looking at the titles and abstracts of reputable papers were looked over to determine those that were important to the study. It was decided that 120 good papers were not relevant to the study because they did not use ensemble learning in disease prediction or any of its different methods for different disease prediction uses.

The second screening resulted in 132 relevant papers for full-text review. Figure 15 depicts the distribution of these selected papers across various disease prediction areas,

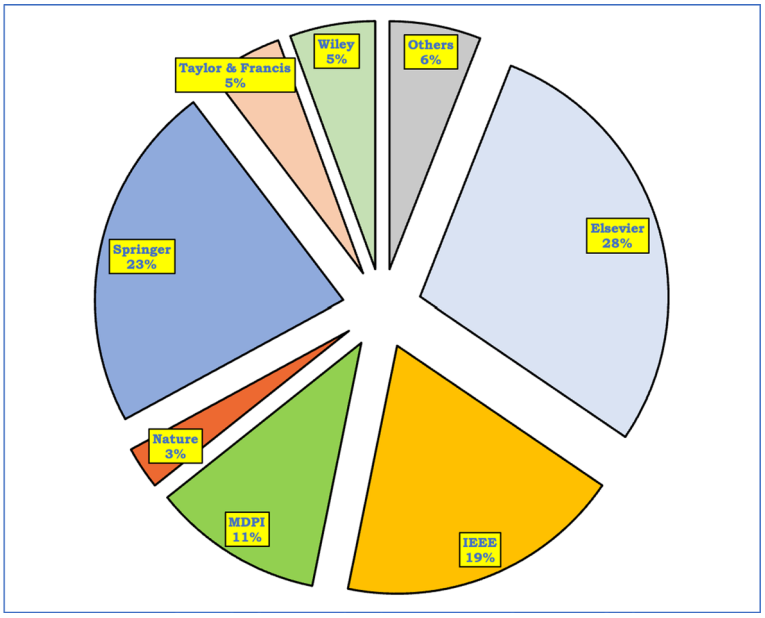


Fig. 13 Percentage of publication from well known publishers

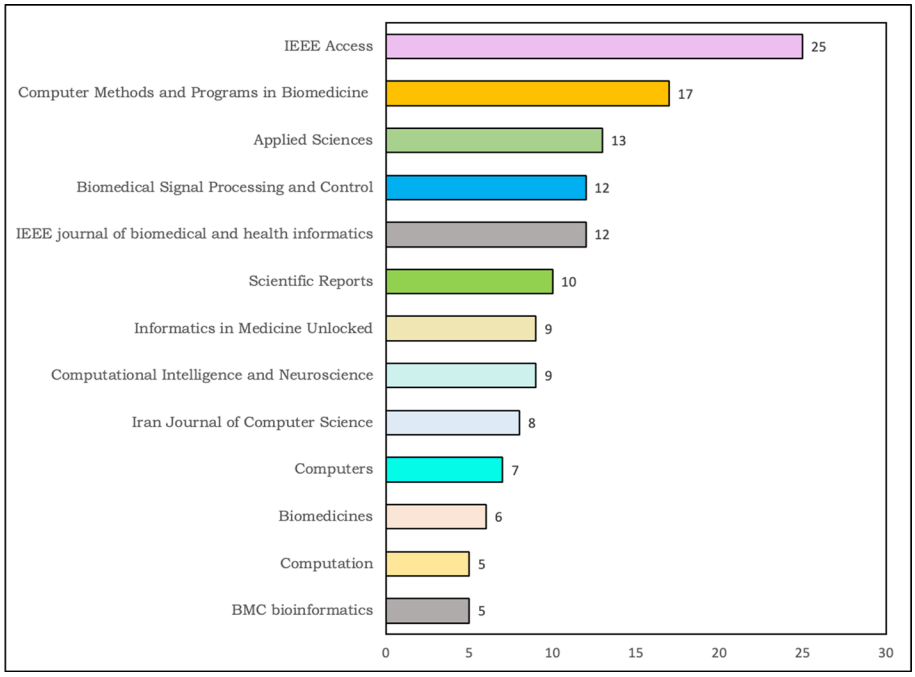


Fig. 14 Number of paper published in various journals

focusing on major diseases prediction such as heart, diabetes, liver, kidney, and skin disease. Among these 132 publications, skin disease prediction accounted for the highest percentage, representing 26%.

In the final screening phase, 132 relevant papers were evaluated against the inclusion criteria outlined in the methodology. The method used makes sure that only investigations of the highest caliber have been taken into account. Consequently, 87 publications were included for additional analysis after meeting the eligibility requirements.

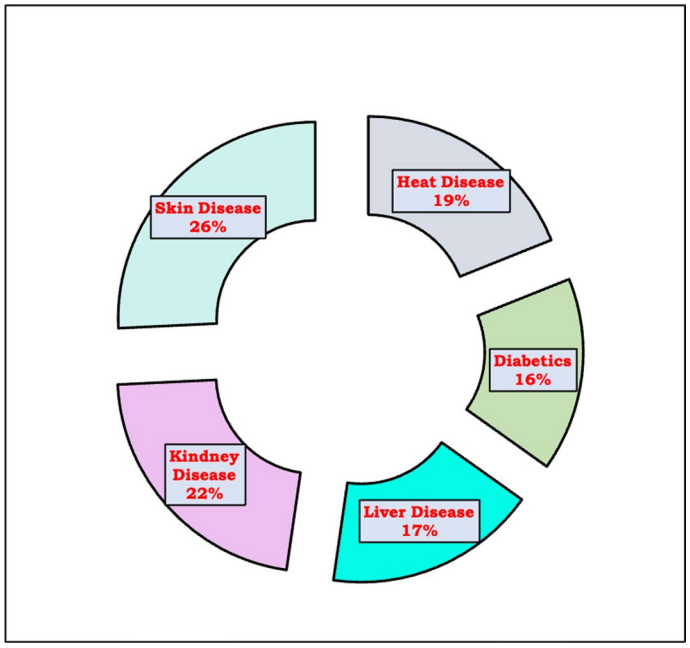


Fig. 15 Percentage of publication used ensemble learning in major diseases prediction

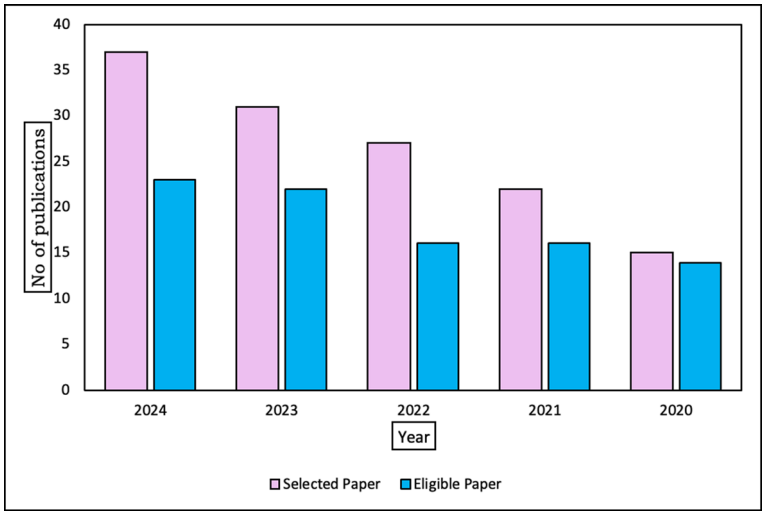


Fig. 16 No of eligible paper and selected paper from 2020 to end of 2024

Additionally, these 87 papers the reference lists were reviewed, leading to the identification of four more studies that satisfied the inclusion criteria. As a result, there were 91 publications in the final dataset.

Figure 16 shows the current trend in the publication of eligible and selected papers on using ensemble learning to predict diseases from 2020 to the end of 2024. In this case, pink colour indicated the number of selected papers (132), and blue indicated the number of chosen or eligible papers (91). It is clear that the trend is going up every year.

In the reporting phase, the papers that were chosen as suitable for the evaluation phase are reviewed again so that we can analyse them further. Figure 17 displays the percentage of selected papers over the last five years, that is, from from 2020 to 2024. In the final set of papers, for heart disease, there are 20% of papers; for diabetics, there are 18%

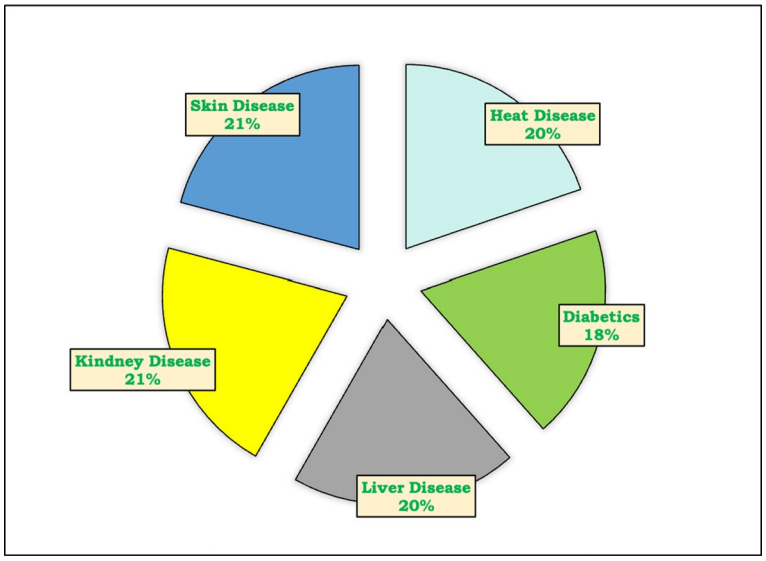


Fig. 17 Percentage of selected paper of different disease from 2020 to end of 2024

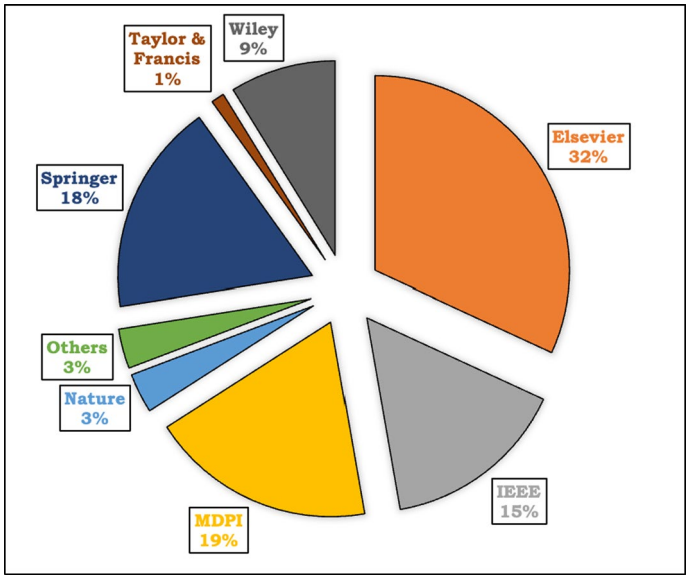


Fig. 18 Percentage of selected publishers of different disease from 2020 to end of 2024

of papers; for liver disease, there are 20% of papers; for kidney disease, there are 21% of papers; and for skin disease, there are also 21% of papers (Fig. 18).

Fron Fig. 21 distribution of these eligible papers, showing the reputable publishers that put them out there. Here, depicted, the majority of these 91 papers were published by Elsevier (32%), i.e., 29 papers and MDPI (19%) i.e., 17 papers, Then follow by Springer and IEEE publisher, contributing 16 and 14 papers, respectively.

Figure 19 also shows eligible papers that have been published in different reputable journals. During the review process, we selected 91 papers, eleven of which were published in IEEE Access and secured the first position. The following seven journal articles placed second in Computer Methods and Programming in Biomedicine, and then six were published in Biomedical Signal Processing and Control and got a third place. The

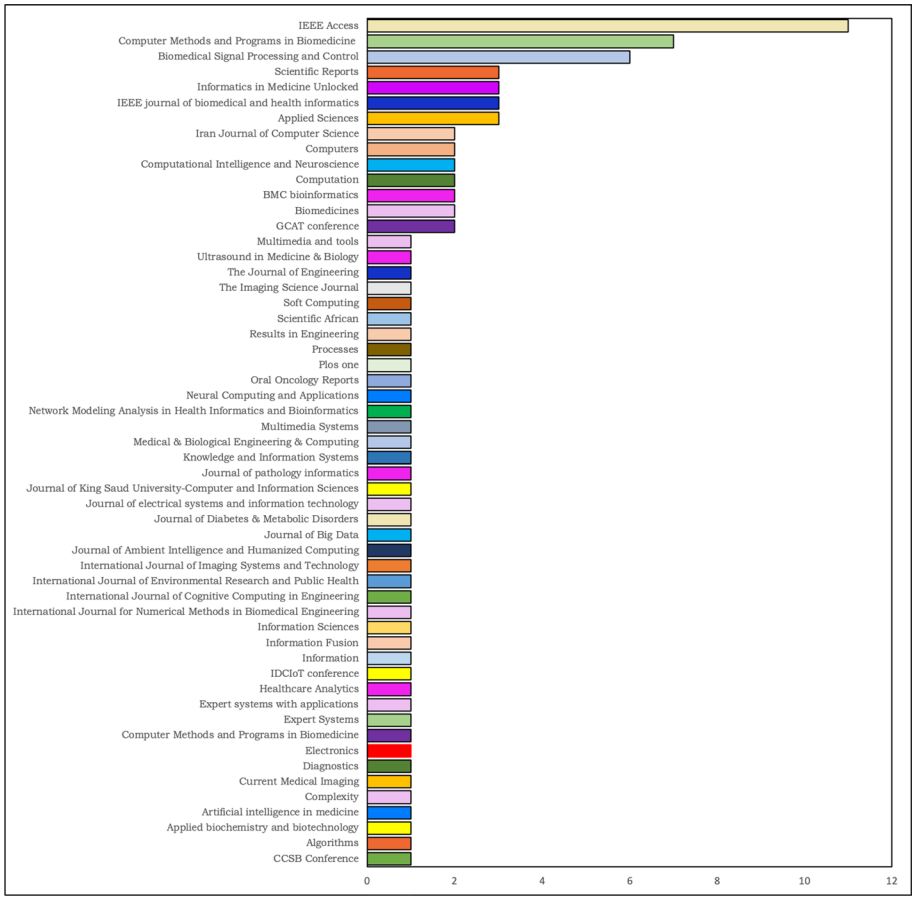


Fig. 19 Eligible paper published in reputed journal

IEEE Journal of Biomedical and Health Informatics, Informatics in Medicine Unlocked, and Scientific Reports were the next three publications.

Figure 20 illustrated the distribution of eligible papers by country. The figure highlights the number of eligible papers that apply disease prediction using ensemble larning. This figure show that most of the paper published in India, followed by China.

This study included the PRISMA approach, that provided several significant advantages. It allows a comprehensive and systematic examination of the existing literature through the methodical exploration of several databases. This comprehensive search methodology reduces a possibility of publication bias and ensures a full representation of the literature. Secondly, by specifying specific inclusion and exclusion criteria, we verified that the chosen publications were directly pertinent to the study's core aim. The systematic review method enabled the execution of a meta-analysis, providing a quantitative summary of the data and increasing the study's statistical power.

Despite its advantages, this systematic review possesses significant drawbacks. Initially, it exclusively depends on literature published in peer-reviewed journals, disregarding unpublished studies, such as preprints, that could offer significant insights. Secondly, because language bias may be introduced, studies published in languages other than English were excluded due to language limitations. The quality and limits of the included research varied, potentially affecting the overall robustness of the conclusions. Moreover, the criteria for inclusion and exclusion, data extraction, and quality evaluations

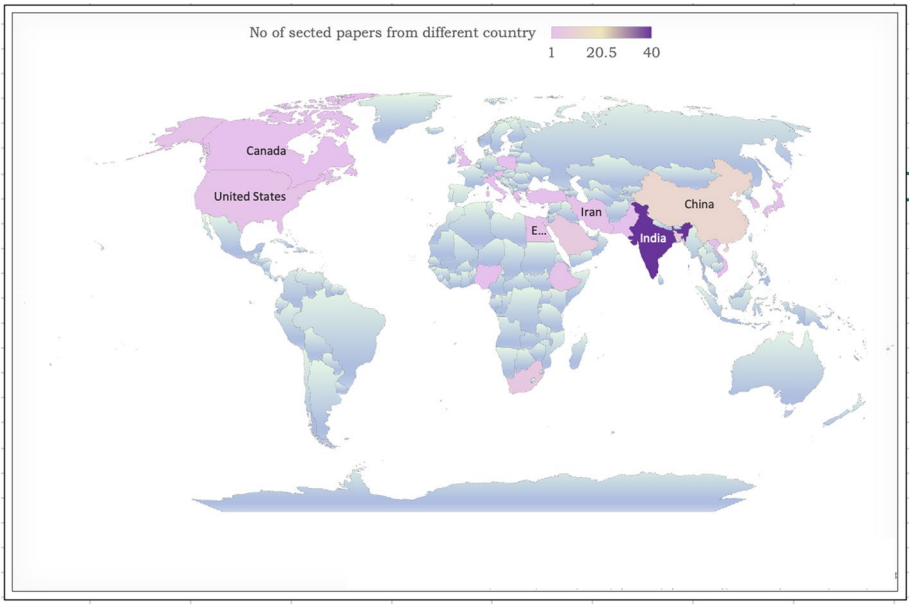


Fig. 20 No of eligible paper from different countries

Table 8 Statistical summary of dominant ensemble learning trends and promising hybrid models across disease domains

Disease domain	Dominant ensemble type	Approx. % studies using stacking/hybrid	Feature selection usage (%)	Accuracy range	Most promising hybrid models
Heart disease	Stacking and hybrid ML ensembles	∼42%	∼55%	94–97%	Stacking (XGBoost–RF–SVM), CNN–LSTM Hybrid, PSO-based Autoencoder
Diabetes	Stacking and weighted ensembles	∼45%	∼60%	93–99%	Hybrid Stacking (SVM–kNN–RF), Deep Ensemble (RF–DT–NN–XGB), GA-based Hybrid Models
Liver disease	Stacking and boosting-based ensembles	∼40%	∼58%	92–99%	Boosting + Feature Selection (XGB/LGBM), DL–ML Hybrids (CNN–SVM), RF-based Ensembles
Kidney disease	Hybrid deep and boosting ensembles	∼48%	∼70%	95–100%	RFE + LightGBM, Stacking (RNN/LSTM/GRU–SVM), Deep Autoencoder Ensembles
Skin disease	Deep hybrid ensembles	∼46%	∼65%	92–99%	CNN–ResNet–DenseNet Stacking, Optimized SVM–RF Ensembles, Explainable CNN Ensembles

were dependent on the authors' subjective evaluations, potentially impacting the review process.

5.1 Cross-disease insights and challenges

While exploring heart disease related ensemble models, a consistent pattern has been noticed that stacking-based and hybrid ensemble models combined with feature selection consistently outperform bagging methods and single classifiers for heart disease related classification approach as shown in Table 8. For diabetes prediction, again a consistent trend is revealed where stacking, hybrid and weighted ensemble models combined with feature selection achieve highest and balanced performance on different

datasets. Stacking based approaches which integrates heterogeneous classifiers reported more than 95% accuracy. While, soft/max voting based approaches without feature selection shows lower accuracy or imbalanced metrics. Similarly, for liver related diseases stacking, boosting and heterogeneous ensemble models offering feature selection achieved highest and most stable predictive performance. These methods reported very high accuracy and balanced precision, recall, and F1-scores compared to voting and basic bagging based approaches. for kidney disease, stacking, hybrid and boosting-based ensemble models which integrated deep learning or gradient boosting with feature selection achieved the highest predictive performance, however the major limitation of these approaches is high computational complexity. While simple models based on bagging or voting shows lower accuracy. While reviewing skin disease prediction studies it is noticed that stacking, hybrid deep ensembles with optimization frameworks outperformed simple voting or standalone model particularly on benchmark datasets such as HAM10000 and ISIC. Overall, while feature-selected stacking and hybrid ensembles emerge as the most effective and generalizable solutions across medical diagnosis tasks. Overall, 40% of the studies employed Stacking and Hybrid ensembles model with feature selection while the the least used technique is voting and bagging based models as shown in Fig. 21.

One of the major challenges noticed across all five disease domains, models are mostly trained on benchmark or single-institution datasets which is limited to demographic diversity. As a results models trained on these data dominated by a single ethnicity, age group may fail when applied to underrepresented populations. For example, heart and kidney disease risk factors across ethnic groups while skin disease models trained on light skin tones may perform incorrectly on darker skin tones. hence, these ensemble hybrid models should be trained with multi-dataset and cross-population validation. Another major challenge is the computational complexity and practical deployment of the hybrid deep ensemble models. These models involves high computational cost for the training which might limit the real-time deployment especially in kidney and heart disease monitoring systems. Hence, lightweight frameworks with shallow meta-learners

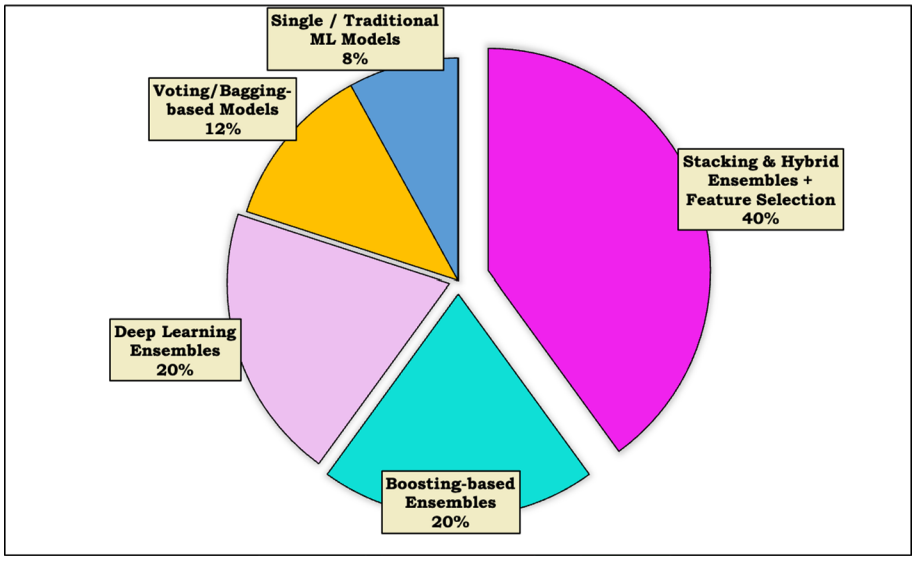


Fig. 21 Dominant modeling patterns observed across heart, diabetes, kidney, liver, and skin disease prediction studies

can be used which requires lesser parameters to train the hybrid models. Also, these black-box ensemble models reduces clinical trust. Integration of explainable AI may help feature importance driven clinical rule validation.

5.2 Explainable AI integration with ensemble learning

Ensemble learning models can achieve strong predictive performance for heart, diabetes, kidney, liver, and skin disease prediction tasks. However, these models are not clinically adopted because of their limited interpretability. Here, Explainable Artificial Intelligence (XAI) techniques can be integrated with the ensemble learning frameworks to ensure transparency, clinical trust, and decision support. For heart and kidney related diseases, XAI models such as SHAP and LIME can be employed to interpret ensemble outputs by quantifying feature importance at both global and individual patient levels. These models may help healthcare professionals to validate the ensemble predictions align with established medical knowledge, such as the impact of blood pressure, cholesterol levels, or creatinine values. In stacking based models, SHAP is often applied at the meta-learner level. For disease related to diabetes and liver function, XAI techniques can be used to address class imbalance and dataset heterogeneity. Feature attribution methods help the identification of dominant risk factors. In skin disease prediction, visualization based XAI models such as Grad-CAM and saliency maps can be combined with the deep hybrid ensembles involving CNNs and ResNet architectures. These models can highlight discriminative regions in the clinical images. This might help the doctors to verify whether ensemble models focus on medically relevant lesion structures rather than spurious artifacts. However, applying these XAI models to the complex stacking and hybrid deep ensemble architectures may cause increased computational overhead. As a result real-time inference may be delayed which can reduce the clinical usefulness and may compromise patient safety.

6 Conclusion and future direction

This systematic review is mostly about studies that look at how different ensemble learning approaches. It can be used to predict major diseases and have been published in reputable journals. Using PRISMA framework [49], the study followed three crucial steps: reporting, evaluation, and identification. During the identification phase, we obtained a preliminary pool of 287 documents from various databases and sources using specific keywords (described in Sect. 3) and search queries. During the review phase, 91 eligible papers for in-depth study were selected from the pool using a strict three-step screening procedure. We meticulously crafted the inclusion criteria to ensure the selection of pertinent and superior studies that explicitly employ ensemble learning techniques for disease prediction. We methodically analysed and synthesised this research throughout the reporting phase to identify key trends and themes. The use of ensemble learning in the prediction of several diseases, including diabetes, skin, kidney, liver, and heart disease, is covered in the review.

The review analyses showed that ensemble learning shows considerable potential in addressing disease prediction issues. Of the 287 papers discovered on this topic from 2020 to December 2024, a significant plurality, 95 papers, were published in 2023; however, only 37 papers were selected for our review, and merely 23 papers are deemed suitable. In contrast, 2022 witnessed the release of 22 qualifying papers, exceeding the

number published in 2023 and 2024. Elsevier, MDPI and Springer were leading publishers, providing the majority of qualifying papers in the responsibility.

The results showed that ensemble learning is a simple to understand effective approach to addressing prediction challenges in multiple diseases, such as diabetes, liver, kidney, heart disease, and skin diseases. Researchers have developed many ensemble learning models for disease prediction purposes. Important strategies for constructing ensemble models for predicting diseases encompass algorithm hybridisation, parameter control modification, and improvement of the prediction strategy. The review findings indicated that ensemble learning has been predominantly utilised in forecasting cardiac disease, with 28 qualifying papers published on this subject in comparison to others. Indian scholars have made more contributions than researchers from any other country to the use of ensemble learning for disease prediction; 42 of their articles are eligible.

We present an in-depth and current examination of the effects of several ensemble learning methods and parameters on illness prediction tasks, including the prediction of diabetes, liver, kidney, skin, and heart diseases. Additionally, we review hybrid and adaptive ensemble learning methods that can adjust to the dynamic and uncertain nature of disease data, such as variations in patient characteristics, medical history, and environmental factors. Our findings reveal that approximately 40–45% of the reviewed studies across heart, diabetes, liver, kidney, and skin disease prediction used stacking or hybrid ensemble models where 60–65% studies incorporated feature selection strategies.Also, it is noticed that most of the feature seelction based hybrid models obtained more than 95% accuracy. It is also observed that voting and bagging-based approaches without applying feature selection techniques are the least used. These statistical findings substantiate that stacking-based and hybrid ensemble frameworks represent the most popular effective models in current disease prediction research. This study recommends the following directions for future research.

First, Adaptive ensemble learning models should be developed and studied in future for dynamical adaptation to variations in patient characteristics, medical histories, and environmental factors. These models will improve illness prediction models' adaptability and robustness across a range of demographics.

Second , Though ensemble learning has been widely applied in cardiac, diabetes, liver, kidney, and skin disease prediction, further studies should focus on its application in less-studied or emerging diseases, like neurodegenerative disorders and infectious diseases.

Third, Future research should move beyond single-dataset evaluations and focus on multi-center, multi-demographic and cross-population datasets. This can train and validate ensemble models on geographically and ethnically diverse cohorts.

Fourth, Rather than relying on post-hoc explanations, future studies should focus on system integration of lightweight XAI models with optimized ensemble models.

Fifth, The increased health monitoring with IoT devices opens opportunities for ensemble learning in real-time prediction of diseases. Future studies are needed to design methods to merge continuous streams from wearable devices with predictive models.

These recommendations can act as a framework for enhancing ensemble learning methodologies and their use in the swiftly progressing domain of disease prediction.

Appendix A Ensemble learning: concepts, algorithms

The fundamental structure of any ensemble learning system entails amalgamating predictions from a collection of baseline classifiers $c_1, c_2, \ldots, c_h$ through an aggregation function G to get a singular output. Consider a dataset D consisting of n samples, where each sample is represented as (x_i, y_i) for $i = 1, 2, \ldots, n$. The input features x_i belong to an m -dimensional real-valued space, denoted as $x_i \in \mathbb{R}^m$, and y_i represents the corresponding target or output variable, the prediction of the ensemble approach can be mathematically articulated as Eq. A1.

$$\hat{y} = G(c_1(x), c_2(x), \ldots, c_h(x)) \tag{A1}$$

where, the expected output is $\hat{y}$,the input feature vector is x, and G is denoted the aggregation function that combines the outputs of the basic classifiers $c_1, c_2, \ldots, c_h$. The selection of G (e.g., majority voting, weighted averaging, or stacking) is contingent on the specific job and the heterogeneity of the base classifiers. This approach establishes the basis for improving predictive accuracy by utilizing the combined advantages of individual classifiers.

Figure 22 depicts the overarching abstract framework of ensemble learning. It illustrates the training of a set of baseline classifiers, collectively known as a classifier ensemble, on input data. These classifiers yield separate predictions, which are subsequently amalgamated using an aggregation method to form a singular, composite prediction [142]. Ensemble techniques differ in their selection of baseline classifiers for training. Two primary methodologies produce variety among base classifiers: homogeneous ensembles, using classifiers of identical types, and heterogeneous ensembles, integrating classifiers of various types shown in Fig. 23 [143].

Homogeneous ensembles [144, 145] consist of baseline classifiers of same type, each trained on distinct data subsets. This strategy maintains a uniform feature selection process throughout the training datasets. The main problem with homogenous ensembles is

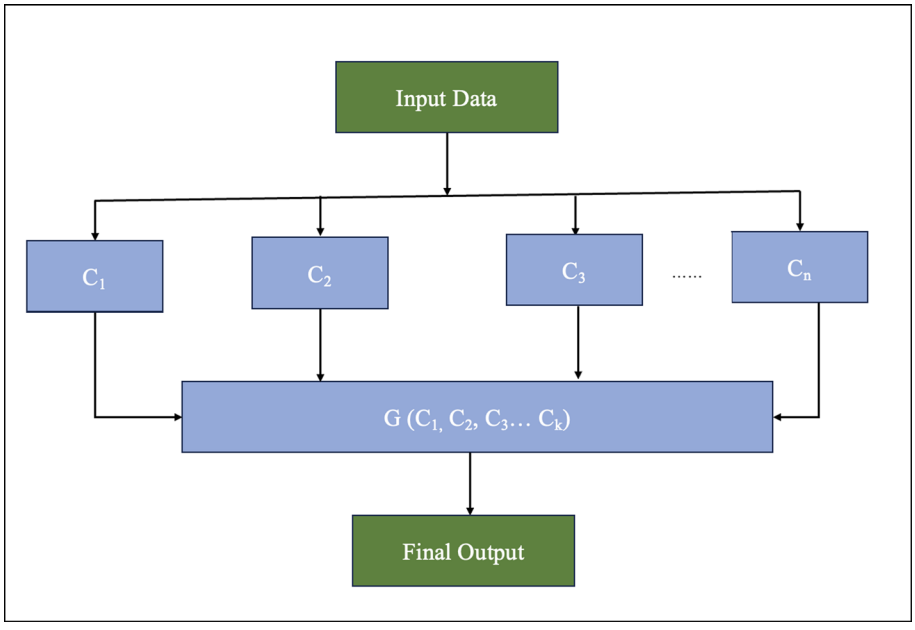


Fig. 22 General framework of the ensemble learning

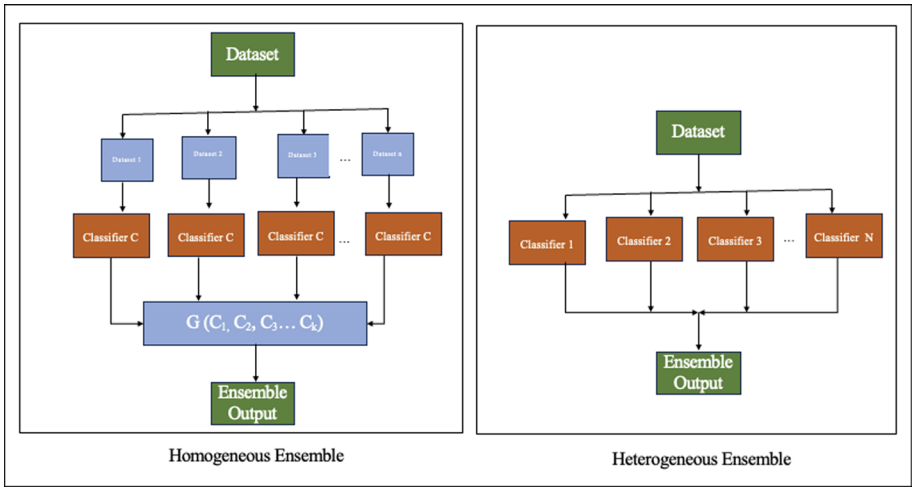


Fig. 23 Homogeneous and heterogeneous ensemble learning framework

Table 9 Characteristics of popular ensemble methods

Method	Dependent	Fusion method	Heterogeneity
Bagging	Parallel	Weight voting	Homogeneous
Random forest	Parallel	Weight voting	Homogeneous
Boosting	Sequential	Weight voting	Homogeneous
AdaBoost	Sequential	Weight voting	Homogeneous
Gradient boosting	Sequential	Weight voting	Homogeneous
Extreme gradient boosting	Sequential	Weight voting	Homogeneous
Stacking	Parallel	Meta learning	Heterogeneous
Hybrid Ensemble	Both	Both	Heterogeneous/homogeneous

producing adequate variability while employing the same learning procedure. Conversely, heterogeneous ensembles [146, 147] have diverse baseline classifiers, each trained on the identical dataset. Diversity is achieved by utilizing various feature selection techniques on the identical training data. Homogeneous ensembles are frequently more attractive to researchers because of their simplicity, ease of implementation, and reduced cost in comparison to heterogeneous ensembles [148].

In general, they can describe and rate an ensemble framework based on three main factors that affect the way it works. The main thing that makes them unique is the dependency structure of the basic models, which can be either sequential or parallel. The second method is called "fusion," and it involves choosing the right way to combine the results of the basic classifiers. Some examples are weighted voting and meta-learning. The third thing that sets uniform and heterogeneous ensembles apart is the heterogeneity of the baseline classifiers. Table 9 illustrted these properties across popular ensemble methods, which will be further upon in the subsequent sections.

A.1 Data sampling technique

Identifying a data sampling method is one of the most important factors that affects how well an ensemble system works. It is important to provide variation in the data sampling selections for the baseline classifiers. In ensemble systems, two primary methods for selecting samples from the training dataset are the independent dataset strategy and the dependent dataset approach [149].

In the independent datasets strategy [150, 151], the subsets are autonomous and do not depend on each other. In contrast, the dependent datasets technique [152, 153] includes interdependent subsets, where each subset is impacted by the outcomes of the previous subset. The main advantage of the independent datasets method is that the performance of one subset remains unaffected by others, whereas with the dependent datasets strategy, the results of previous subsets directly affect subsequent ones.

Both approaches face challenges, such as deciding the appropriate size of each data sample and determining the maximum number of samples that can be used effectively. Moreover, the selection of an effective data sampling approach is contingent upon the specific ensemble method employed [30, 154–156]. These factors are essential to improving the general efficacy of the ensemble method.

A.2 Baseline classifiers development

The effectiveness of ensemble systems is significantly influenced by the diversity of baseline classifiers. The sequential ensemble method and the parallel ensemble method are the two main techniques that ensemble methods often use to train individual components [157, 158].

In the sequential ensemble technique [159, 160], learners are trained sequentially due to data dependency. Each model consecutively rectifies the inaccuracies of its predecessor, as demonstrated in Fig. 24. This method depends on the interdependence among base learners, that constitutes its primary advantage [161].

Conversely, the parallel ensemble technique [162] produces base learners concurrently, as there exists no dependence among the data. Each base learner is trained independently, as illustrated in Fig. 25. The key advantage of this approach lies in leveraging the independence of base learners. As a result, the errors generated by one model differ from those of others, enabling the ensemble model to effectively mitigate errors through averaging [163, 164].

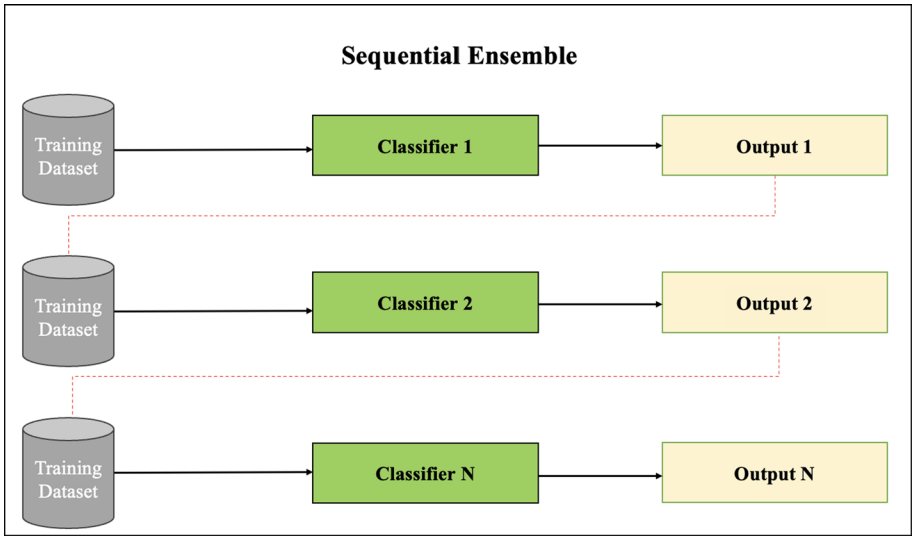


Fig. 24 Sequential ensemble learning framework

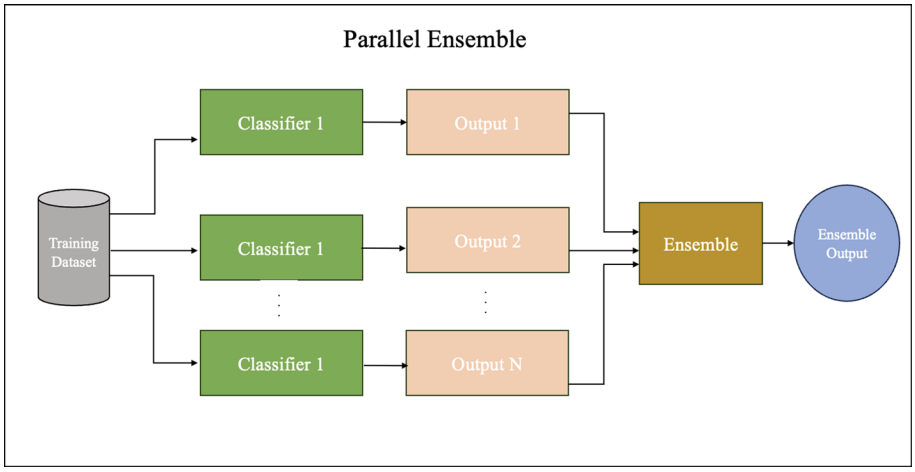


Fig. 25 Parallel ensemble learning framework

A.3 Fusion method in ensemble learning

The fusion analysis in ensemble machine learning demonstrates ways to integrate the results or predictions from various base models to provide a compelling result. Fusion techniques enhance the ensemble's effectiveness by leveraging on the strengths of individual models, reducing errors, and enhancing accuracy, stability, and adaptability. Morover, the fusion in ensemble learning basically includes two techniques: the voting technique and the meta-learning technique [32, 165]. The discussion will delineate the performance of each approach for combining the outputs of baseline classifiers, emphasize their benefits, confront the problems linked to their application, and ascertain the most appropriate fusion method for diverse ensemble approaches. These fusion techniques are flexible and can be utilized with both independent and dependent data samples, as well as with baseline classifiers that are trained in parallel or sequentially.

A.3.1 Types of voting techniques

Voting techniques are widely used in classification and regression tasks to enhance the accuracy and effectiveness of predictions. They are particularly suitable for including models in ensemble methods like as bagging and boosting. The initial fusion method, voting ensembles, consists of three fundamental methods: max voting, average voting, and weighted average voting. Each method will be evaluated for its implementation, benefits, and limitations.

- *Max Voting techniques*: The most commonly utilized voting method is max voting, also referred to as majority voting or hard voting [166]. This method combines the predictions from multiple classifiers and selects the class label with the highest number of votes. The mathematical representation is provided in Eq. A2.

$$y^* = \text{mod}[C_1(x), C_2(x), \ldots, C_n(x)] \tag{A2}$$

Here, y^* represents the class label determined by the majority (or plurality) of votes from the predictions of individual classifiers C_n. For example, consider three classifiers C_1, C_2 and C_3 that predict the following outputs for a given sample: [0, 0, 1]. Using the max voting method, the predicted label is determined as $y = mode[1, 1, 0] = 1$ assigning the sample to "class 1." This method is often utilized in ensemble techniques such as

bagging. A modified version of max voting is soft voting, where the predicted probabilities for each class are combined, and the class with the highest cumulative probability is selected as the final prediction. This approach is mathematically represented in Eq. A3.

$$y^* = \arg\max_i \sum_{j=1}^{n} w_j P_{ij} \tag{A3}$$

In this formula, w_j denotes the weight assigned to the j-th classifier, and P_{ij} represents the predicted probability for class i given by the j-th classifier. The final prediction y^* is the class with the maximum weighted sum of probabilities. In contrast to max voting, which simply accounts for the ultimate prediction of each classifier, soft voting necessitates the retention and application of the complete probability distribution, rendering it both computationally and storage-intensive. Soft voting facilitates adaptability in prediction calculations, including the selection of the class with the highest probability or the averaging of probabilities among classifiers [166]. However, max voting is beneficial due to its simplicity and ease of application, providing it the most simple voting system. Nonetheless, its disadvantages encompass the computational expense linked to using several baseline models and its inefficacy when all classifiers yield identical results. Moreover, maximum voting may not be appropriate for all categories of problems [167–169].

- *Averaging Voting*: Averaging votes is the second voting technique [170, 171]. This method entails integrating predictions from various models and calculating the average of these estimates to arrive at the final prediction. In order to get the average, the arithmetic mean, defined as the total of all predictions divided by the number of predictions, shown in Eq. A4.

$$y^* = \arg\max_i \frac{1}{n} \sum_{j=1}^{m} w_{ij} \tag{A4}$$

Where w_{ij} is the probability of the i-th class label predicted by the j-th classifier. Consider an ensemble with three classifiers producing the following predicted probabilities for two classes:

$$C_1(x) = [0.7, 0.3], \quad C_2(x) = [0.4, 0.6], \quad C_3(x) = [0.8, 0.2].$$

The average probabilities are computed as: For class 0:

$$y_0 = \frac{0.7 + 0.4 + 0.8}{3} = 0.633,$$

For class 1:

$$y_1 = \frac{0.3 + 0.6 + 0.2}{3} = 0.367.$$

Since class 0 has the higher average probability, the final prediction is:

$$y = 0.$$

The average voting method is beneficial because to its robust predictive capability, enhanced accuracy compared to majority voting, and its capacity to reduce overfitting. Furthermore, it functions as a natural substitute for maximum voting in the bagging technique. Nonetheless, average voting possesses specific disadvantages, such as increased computing expense compared to max voting, as it requires the aggregation of predictions from all baseline models. Another limitation is the assumption that each model in the ensemble is equally effective, which may not be accurate, as certain models [25, 172].

- *Weighted Average Voting*: Weighted average voting, a variation of the conventional averaging voting procedure, is the third voting technique [173]. The baseline classifiers are given different weights in this method, demonstrating the significance each model is to the prediction process. To calculate the weighted average, each prediction is multiplied by its respective classifier's weight to produce a weighted sum. The combined total of the weights of all classifiers is then used to divide the resulting amount. Each class, such as class 0 or class 1, has its weighted average determined by the weights, as seen in Eq. A5.

$$y^* = \frac{\sum_{j=1}^{m} w_j x_j}{\sum_{j=1}^{m} w_j} \tag{A5}$$

Where w_j denotes the weights, m represents the number of terms to be averaged, w_j refers to the weights assigned to the values x_j, and x_j are the data values being averaged. For instance, assume the ensemble of classifiers consists of three members:

$$C_1(x) = [85.0, 15.0], \quad C_2(x) = [90.0, 10.0], \quad C_3(x) = [88.0, 12.0],$$

and the assigned weights for the classifiers are:

$$[0.80, 0.85, 0.75].$$

To compute the prediction for class 0:

$$y_0 = \frac{(85.0 \times 0.80) + (90.0 \times 0.85) + (88.0 \times 0.75)}{0.80 + 0.85 + 0.75} = 87.216,$$

and for class 1:

$$y_1 = \frac{(15.0 \times 0.80) + (10.0 \times 0.85) + (12.0 \times 0.75)}{0.80 + 0.85 + 0.75} = 12.784.$$

Since class 0 has the higher weighted average, the final prediction is:

$$y = 0.$$

The weighted average voting method provides enhanced precision relative to the simple average voting methodology. However, its primary issue resides in determining the suitable weights for each ensemble member. Furthermore, it is more computationally demanding than the typical voting approach, as it necessitates the calculation of the weighted average of forecasts from all baseline models. This increased complexity constrains its actual application in certain circumstances [174].

A.3.2 Meta learning technique

Meta-learning technique [175] is the second fusion technique, also known as "learning to learn." This method entails integrating data from multiple learners to obtain insights. Meta-learning modifies certain aspects of a learning algorithm based on trial outcomes, leveraging prior experience with various tasks to increase the program's efficiency. In contrast to conventional ML models, meta-learning operates in many stages, using the outputs of individual models (inducers) as inputs for a more advanced model known as the meta-learner, which generates the final prediction [176, 177].

Interest in meta-learning has markedly increased during the past five years, especially since 2017. The proliferation of powerful ML algorithms has introduced obstacles in model training, resulting in increased focus on meta-learning. ML algorithms encounter various challenges, including substantial operating expenses caused by the necessity of extensive experimentation during the training phase, which can be time-consuming in determining the optimal model for a certain dataset. These challenges are mitigated by meta-learning, which enhances learning algorithms and finds models with better performance [176, 177].

Meta-learning offers advantages such as expediting the learning process by reducing the number of required experiments, enhancing algorithm adaptability to dynamic environments, and modifying hyperparameters for better results. Moreover, meta-learning offers solutions to other ML difficulties, including the small dataset management, the mitigation of computational complications, and the enhancement of generalization. The main challenge with meta-learning is to methodically learn from experience in a data-driven way [178, 179].

Among various meta-learning approaches, stacking is one of the most widely used techniques [44, 180]. Implementing meta-learning presents challenges such as selecting the suitable meta-learning technique, handling computational complexities, and addressing concerns related to big datasets, various baseline models, or hierarchical meta-learning layers [180].

A.4 Popular ensemble approaches

Bias and variance are frequent problems for ML models. Bias is defined as a difference between the anticipated values generated by the model and the actual values.High bias models frequently oversimplify the data's underlying patterns, which results in inaccurate predictions for testing and training. significant variance and significant bias are common characteristics of such a model. Therefore, ensemble methods are being developed to improve the model's accuracy. One traditional ML technique is an ensemble technique, that combines weak classifiers and assigns the individual model prediction for the final prediction.

A.4.1 Bagging technique

Leo Breiman first proposed the bagging approach, also known as bootstrap aggregating, in 1994 as a way to enhance ML algorithms' predictions by combining predictions from several randomly chosen training datasets [181]. Breiman showed that good performance improvements in models resulting from small perturbations of the training data were possible by perturbing the original data, thus enabling bagging to be a really effective way for reducing overfitting and maximizing accuracy [181]. The idea of bagging is that

by generating the multiple bootstrapped replicas of the original data, it tries to introduce diversity. This occurs because each copy is made using a random sampling technique that replaces the input data. The same ML technique is utilized to train multiple learners using such distinct subsets.

Bagging operates by creating $'b'$ random subsets of the training data, each used to train an independent base learner. To produce a more reliable and accurate final model, the predictions made by these base learners are then combined, usually by majority vote for classification problems. This process is represented in Algorithm 1.The RF algorithm, which uses the bagging technique to train an ensemble of decision trees, is one of the most widely used applications of this method.

Require: Training set $D = \{(a_1, b_1), (a_2, b_2), \ldots, (a_m, b_m)\}$, learning algorithm A, total models K
Ensure: Final ensemble predictor $H(a)$
1: **for** $k = 1$ to K **do**
2: Create a bootstrap dataset D_k by randomly sampling with replacement from D
3: Train a model f_k using A on D_k, i.e., $f_k = A(D_k)$
4: **end for**
5: Define the ensemble prediction as:
$H(a) = \text{mode}(f_1(a), f_2(a), \ldots, f_K(a))$ (for classification)
$H(a) = \frac{1}{K} \sum_{k=1}^{K} f_k(a)$ (for regression)

Algorithm 1 Bagging ensemble

The major advantage of bagging is over algorithms that show high variance. Algorithms that suffer from high variance are called "unstable" algorithms. Minor modifications to the input data result in significant differences in the model predictions for algorithms such as this one. Examples of unstable algorithms include decision trees, which often overfit; these do exceptionally better when bagged. Algorithms with high stability, such as naïve Bayes and k-nearest neighbors (KNN), don't gain much by bagging, because they are much less sensitive to perturbations in the data [182].

Bagging also has the critical advantage of reducing variation without appreciably increasing bias. Furthermore, the bootstrapping process allows for data variety, which strengthens the model. For large data sets, bagging provides efficiency in computation because training can be performed on smaller sample subsets [183]. The downside of bagging is that its interpretability is reduced. For instance, the ensemble of a number of decision trees in bagging obscures interpretability even though a single tree is highly interpretable and has a clear structure. In addition, some critical features may not be used during training due to random sampling, and thus some potential insights may be missed. However, bagging is still a very powerful technique for improving model accuracy, especially in scenarios where variance reduction is crucial.

- Random Forest (RF)

The RF algorithm, developed by Breiman in 2001 [184], is a very robust ensemble approach that builds on bagging to increase the performance of decision trees. It developed multiple decision trees using bootstrapped data subsets and produced the process unpredictable by choosing a random subset of features to consider at each split. The overall accuracy and adaptability of the model are improved by the decorrelation of these trees.

For classification, RF aggregates the predictions of the individual trees through majority voting. For regression tasks, it averages their outputs. Its ability to handle high-

dimensional data and noisy inputs, as well as its provision of feature importance rankings, makes it very versatile.

An important benefit of RF is its ability to not overfit, even with many trees, and it has excellent generalization ability. However, this comes at the cost of interpretability, because the ensemble of trees is essentially a "black-box" model. Additionally, training and prediction may be computationally expensive, depending on the number of trees or the size of the dataset. However, for both classification and regression tasks, One of the most widely used and exact algorithms is still the RF algorithm. The RF algorithm is presented in Algorithm 2.

Require: Training dataset $D = \{(a_1, b_1), (a_2, b_2), \ldots, (a_m, b_m)\}$, number of trees K, feature subset size F, base learning algorithm A (e.g., decision tree)
Ensure: Final ensemble model $H(a)$
1: **for** $k = 1$ to K **do**
2: Generate a bootstrap sample D_k from D (sample with replacement)
3: Train a decision tree f_k using A on D_k
4: At each node in f_k, randomly select F features (out of total features) and determine the best split using only these F features
5: **end for**
6: Define the final model as:
$H(a) = \text{mode}(f_1(a), f_2(a), \ldots, f_K(a))$ (for classification)
$H(a) = \frac{1}{K} \sum_{k=1}^{K} f_k(a)$ (for regression)

Algorithm 2 Random Forest

A.4.2 Boosting

A ML technique called "boosting" is intended to turn ineffective learners into precise classifiers. It is a composite meta-algorithm used to reduce prediction model volatility and bias. A classifier that performs only slightly better than random guessing is called a weak learner, whereas strong learners achieve high accuracy and support-boosting ensemble approaches [185].

Schapire first introduced the concept of boosting in a 1990 paper in response to a question posed by Kearns and Valiant about the possibility of integrating weak learners to produce a robust learner [186, 187]. Numerous boosting algorithms, such as AdaBoost [188] and XGBoost [189], were developed as a result of Schapire's research, which had a significant impact on ML and statistics [186].

The core idea behind boosting is applying a simple learning algorithm iteratively on changed copies of the input data [67]. Boosting commences with the training of a weak learner on the original dataset, followed by an assessment of its predictions to pinpoint misclassified training examples [190]. The weak learner is retrained on a modified dataset that gives the misclassified occurrences importance, and the misclassified samples are given greater weights. This process is carried out until a specific number of base learners are produced, then a weighted algorithm is used to combine their predictions. Reducing bias over variation is the main focus of boosting [182, 191]. Through the use of additional weight, the boosting improved the attention on misclassified data, compelling the weak learner to focus on these occurrences and so reducing bias across successive iterations. This method provides boosting susceptible to noisy data. Noisy samples, which are frequently challenging to classify, tend to be assigned disproportionately high weights, resulting in the algorithm concentrating overly on them and perhaps causing overfitting. Notwithstanding this constraint, boosting-based ensemble techniques,

including AdaBoost and Gradient Boosting, continue to be among the most efficacious algorithms in practical ML [192, 193].

- Adaboost

An inventive boosting method called the AdaBoost algorithm combines weak learners to create a powerful classifier. Conceived in 1995 by Freund and Schapire, among the most effective ML algorithms, it is widely considered. The first successful application of a boosting approach was AdaBoost, utilising base learners that are generally single split decession tree, commonly known as decision stumps due to their simplicity. AdaBoost is a notable implementation of AdaBoost, specifically developed for binary classification applications. AdaBoost incrementally modifies the weights of training data, prioritising misclassified instances. This enables the construction of a robust ensemble classifier by directing subsequent weak learners towards the more difficult elements of the dataset. [194] The algorithm's efficacy and adaptability have established it as a fundamental method in ensemble learning. The AdaBoost learning method [195] entails training a foundational classifier utilising a primary algorithm, commonly a decision tree. At the outset, all training samples are allocated identical weights. The predictions are evaluated once the initial classifier has been trained, and the sample weights are adjusted based on the classifier's performance. Correctly identified instances are given lower weights, whereas incorrectly classified examples are given higher weights. This guarantees that succeeding classifiers concentrate more on the difficult, misclassified instances. AdaBoost develops a robust final classifier by iteratively modifying sample weights and training new classifiers, therefore reducing the drawbacks of individual base learners [196]. The AdaBoost algorithm builds an effective classifier by incrementally integrating base learners and assigning weights according on its efficacy. To bring out the more challenging examples that were incorrectly identified in previous rounds, the algorithm adjusts the weights of the training instances at each iteration.

Consider a binary classification task with a labeled dataset containing m instances:

$$D = \{(a_1, b_1), (a_2, b_2), \ldots, (a_m, b_m)\}$$

where $b_i \in \{-1, +1\}$ is the true class label for input a_i.

Initially, each sample a_i is assigned an equal weight in the distribution w_1, defined as:

$$w_1(i) = \frac{1}{m}, \quad \text{for } i = 1, 2, \ldots, m \tag{A6}$$

At each boosting round, the weights are adjusted to emphasize harder-to-classify instances. The updated weight distribution w_{t+1} is given by:

$$w_{t+1}(i) = \frac{w_t(i)}{Z_t} \exp\left(-\beta_t b_i g_t(a_i)\right) \tag{A7}$$

Here, g_t denotes the weak learner trained at iteration t, β_t is its associated confidence score, and Z_t is a normalization term ensuring that w_{t+1} sums to 1.

The values of Z_t and β_t are calculated as:

$$Z_t = \sum_{i=1}^{m} w_t(i) \exp\left(-\beta_t b_i g_t(a_i)\right) \tag{A8}$$

$$\beta_t = \frac{1}{2} \ln\left(\frac{1-\varepsilon_t}{\varepsilon_t}\right) \tag{A9}$$

The classification error ε_t of the learner g_t under the current weight distribution is computed as:

$$\varepsilon_t = \sum_{i=1}^{m} w_t(i) \cdot \mathbb{I}[g_t(a_i) \neq b_i] \tag{A10}$$

After completing all T boosting rounds, the final prediction function $G(a)$ is formed by a weighted combination of all the weak learners:

$$G(a) = \text{sign}\left(\sum_{t=1}^{T} \beta_t g_t(a)\right) \tag{A11}$$

Require: Labeled dataset $D = \{(a_1, b_1), \ldots, (a_m, b_m)\}$, weak learner A, number of iterations T
Ensure: Final classifier $G(a)$
1: Initialize weights: $w_1(i) = \frac{1}{m}$, for $i = 1, 2, \ldots, m$
2: **for** $t = 1$ to T **do**
3: Train weak learner g_t using distribution w_t
4: Compute error: $\varepsilon_t = \sum_{i=1}^{m} w_t(i) \cdot \mathbb{I}[g_t(a_i) \neq b_i]$
5: Set learner weight: $\beta_t = \frac{1}{2} \ln\left(\frac{1-\varepsilon_t}{\varepsilon_t}\right)$
6: Update weights:
$w_{t+1}(i) = \frac{w_t(i)}{Z_t} \exp\left(-\beta_t b_i g_t(a_i)\right)$
where Z_t normalizes w_{t+1} to form a valid distribution
7: **end for**
8: Output the final prediction function:
$G(a) = \text{sign}\left(\sum_{t=1}^{T} \beta_t g_t(a)\right)$

Algorithm 3 AdaBoost

In algorithm 3, provides a summary of the AdaBoost algorithm. Its implementation is relatively simple and necessitates minimal hyperparameter adjustment, rendering it user-friendly and broadly applicable [197]. AdaBoost exhibits considerable flexibility, as it can utilise diverse algorithms as foundational learners. This adaptability permits the selection of an optimal base learner for a particular application, and AdaBoost can further augment its efficacy. Nonetheless, a significant drawback of AdaBoost is its susceptibility to noisy data and outliers. The iterative learning method assigns greater weights to misclassified cases in successive iterations, potentially resulting in overfitting when the dataset includes noise or substantial outliers.

- Gradient Boosting (GB) technique

The GB is an advanced M method that uses boosting to build strong ensembles. As base learners, Gradient Boosted Decision Trees (GBDT) generate a robust ensemble classifier. Breiman [26] proposed gradient boosting as an optimisation approach for a loss function. Friedman [194] extended the gradient boosting technique to expand this approach. Gradient boosting trains models sequentially to improve on earlier errors. Gradient boosting is comparable to other boosting methods, but its base

learners line closely with the ensemble's negative loss function gradient. Iterative, gradient-based optimisation assures the ensemble model minimises loss function and optimises prediction performance [198].

Let us consider a training dataset $T = \{(u_j, v_j)\}_{j=1}^{M}$, where u_j represents the input features and v_j denotes the target labels. The objective of Gradient Boosting is to approximate a target function $G^*(u)$ by minimizing a chosen differentiable loss function $L(v, G(u))$, which evaluates the difference between predicted and actual outputs.

Gradient Boosting builds an additive model $G_t(u)$ incrementally:

$$G_t(u) = G_{t-1}(u) + \eta_t f_t(u) \tag{A12}$$

Here, $f_t(u)$ is the weak learner (e.g., a decision tree) at iteration t, and η_t is the step size.

The model initialization is performed by:

$$G_0(u) = \arg\min_{\gamma} \sum_{j=1}^{M} L(v_j, \gamma) \tag{A13}$$

At each step, the new weak model is chosen to minimize the overall loss:

$$(\eta_t f_t(u)) = \arg\min_{\eta, f} \sum_{j=1}^{M} L\left(v_j, G_{t-1}(u_j) + \eta f(u_j)\right) \tag{A14}$$

Rather than solving this directly, we compute the pseudo-residuals:

$$s_{tj} = -\left[\frac{\partial L(v_j, G(u))}{\partial G(u)}\right]_{G(u)=G_{t-1}(u)}, \quad j = 1, \ldots, M \tag{A15}$$

A new dataset $R_t = \{(u_j, s_{tj})\}_{j=1}^{M}$ is formed, and a weak learner $f_t(u)$ is trained on this set. Then, the best step size η_t is found:

$$\eta_t = \arg\min_{\eta} \sum_{j=1}^{M} L(v_j, G_{t-1}(u_j) + \eta f_t(u_j)) \tag{A16}$$

To reduce overfitting, shrinkage is applied:

$$G_t(u) = G_{t-1}(u) + \lambda \eta_t f_t(u)$$

where $\lambda \in (0, 1]$ is the shrinkage factor that controls the contribution of each learner. The gradient boost algorithm repsented in the algorithm 4.

1: Initialize the model:
$G_0(u) = \arg\min_\gamma \sum_{j=1}^{M} L(v_j, \gamma)$
2: **for** each iteration $t = 1$ to T **do**
3: (a) Compute pseudo-residuals:
$s_{tj} = -\left[\frac{\partial L(v_j, G(u))}{\partial G(u)}\right]_{G(u)=G_{t-1}(u)}, \quad j = 1, \ldots, M$
4: (b) Train weak model $f_t(u)$ on $\{(u_j, s_{tj})\}_{j=1}^{M}$
5: (c) Find optimal step size:
$\eta_t = \arg\min_\eta \sum_{j=1}^{M} L\left(v_j, G_{t-1}(u_j) + \eta f_t(u_j)\right)$
6: (d) Update the model:
$G_t(u) = G_{t-1}(u) + \eta_t f_t(u)$
7: **end for**
8: **return** Final model $G_T(u)$

Algorithm 4 Gradient Boosting

Like other boosting algorithms, GB's primary benefit is its capacity for identifying complex patterns in the input data. This is accomplished by progressively training models to correct the inaccuracies of their forerunners. [199] Gradient boosting models are prone to overfitting and may assimilate noise when the input data is contaminated. Notwithstanding this, the technique is especially appropriate for applications that utilise limited datasets [200].

- XGBoost

An ensemble method for decision trees that makes use of gradient boosting is called XGBoost. The scalable and accurate technique is commonly used for classification and regression. After receiving several Kaggle competitions, XGBoost has become a widely applied ML technique. In 2016, Chen and Guestrin developed [189], it improves gradient boosting methods. A regularisation term in the loss function reduces overfitting shown in Eq. A17 and [201].

$$L_M(F(x_i)) = \sum_{i=1}^{n} L(y_i, F(x_i)) + \sum_{m=1}^{M} \Omega(h_m) \tag{A17}$$

In this case, $F(x_i)$ indicates the forecast for the i-th instance at the M-th iteration. L(*) is a loss function that calculates the difference between the expected and actual target values. The regularisation term $\Omega(h_m)$ controls model complexity and prevents overfitting. Forming the regularisation term as equation A18.

$$\Omega(h) = \gamma T + \frac{1}{2}\lambda\|\omega\|^2 \tag{A18}$$

Where, γ serves as the complexity parameter, controlling the minimal loss reduction necessary for internal node splitting. Higher values of γ result in simpler trees.T indicates the number leaves there are on the tree overall.The penalty parameter is represented by λ. the output values of the leaf nodes is denoted by ω corresponds. Therefor, the modified Eq. A17 written as Eq. A19.

$$L_M \approx \sum_{i=1}^{n} \left[g_i f_m(x_i) + \frac{1}{2} h_i f_m^2(x_i) \right] + \Omega(h_m) \tag{A19}$$

Where, I_j represents the set of samples in the j-th leaf node. g_i the first-order gradient of the loss function with respect to the prediction. The loss function's second-order

gradient (Hessian) is indicated by h_i. A leaf weight regularisation parameter is denoted by λ. The complexity parameter that penalises the number of leaf nodes T is γ. T represents the total number of leaf nodes in the tree. Equation A20 is thus used to express the objection function.

$$L_M = \sum_{j=1}^{T} \left[\left(\sum_{i \in I_j} g_i \right) \omega_j + \frac{1}{2} \left(\sum_{i \in I_j} h_i + \lambda \right) \omega_j^2 \right] + \gamma T \tag{A20}$$

A quadratic approximation of the goal function is used for optimization. Additionally, by adding the regularization term, XGBoost becomes resistant to overfitting [202]. Like the gradient boosting method, XGBoost reduces overfitting by using subsampling, learning rate, and maximum tree depth.

The XGBoost algorithm presents numerous benefits. Since it is naturally capable of handling these circumstances, it requires minimal feature engineering, including data normalization and feature scaling. Furthermore, it can proficiently handle absent values. A significant characteristic of XGBoost is its capacity to provide feature importance, which enhances comprehension of input characteristics and supports feature selection. The approach outperforms most ML algorithms in speed, accommodates big datasets, and exhibits resistance to overfitting. XGBoost frequently surpasses other ML algorithms, which has led to its success in securing multiple victories in Kaggle contests. XGBoost has certain shortcomings, notably its extensive array of hyperparameters, which complicates and prolongs the tuning process [203, 204, 200].

- LightGBM

In 2017, researchers at Microsoft developed the successful gradient boosting method known as Light Gradient Boosting Machine (LightGBM). It is extensively utilised for classification, ranking, and many ML challenges. LightGBM presents two innovative techniques:Exclusive Feature Bundling (EFB) and Gradient-based One-Sided Sampling (GOSS). These techniques allow for faster training while maintaining high accuracy.

The GOSS method enhances the gradient boosting methodology by emphasising training cases with greater gradients, therefore, learning is accelerated and the computational complexity of the model is reduced. The GOSS method neglects a significant quantity of training instances with minimal gradients and uses only the remaining instances to calculate the information gain (IG) [205]. This is justified by the fact that samples with large gradients improve the IG calculation more than samples with small gradients. Consequently, the GOSS approach attains precise IG estimate while utilising a decreased sample quantity [206].

The Exclusive Feature Bundling (EFB) approach conducts feature selection by consolidating sparse, mutually exclusive properties, thus diminishing the feature count [205]. In sparse feature spaces, numerous qualities are nearly exclusive, indicating that they rarely take nonzero values concurrently [206]. An illustrative instance of such characteristics is one-hot encoded attributes. To increase computational performance, the EFB approach reduces the size of the feature matrix by combining these features [205]. The principal benefit of LightGBM is its rapidity,

frequently producing a highly efficient model. Secondly, Since it converts continuous values into discrete bins, it uses less memory.. Thirdly, LightGBM attains greater accuracy compared to the bulk of conventional boosting algorithms, primarily due to the GOSS and EFB methodologies. Moreover, it has superior performance on extensive datasets and demonstrates a reduced training duration relative to the XGBoost algorithm [207]. Nonetheless, LightGBM possesses several drawbacks. It is prone to overfitting short training datasets due to its optimisation for larger datasets. Moreover, its leaf-wise tree-splitting approach can generate more intricate trees, potentially resulting in overfitting.

The LightGBM algorithm has been effectively utilised in numerous classification tasks, yielding outstanding outcomes [208–210]. The approach is detailed in Algorithm 5. For a comprehensive elucidation of the LightGBM methodology, consult [209]. A detailed mathematical analysis of the technique is available in [211].

Require: Training data $S = \{(x_1, y_1), \ldots, (x_n, y_n)\}$, loss function $L(y, \Theta(x))$, number of iterations T, sampling ratios a, b
Ensure: Final model $\hat{\Theta}(x)$
Combine mutually exclusive features using Efficient Feature Bundling (EFB)
Initialize model:
$\Theta_0(x) = \arg\min_c \sum_{i=1}^{N} L(y_i, c)$
for $t = 1$ to T **do**
  Compute gradients g_i and Hessians h_i
  Apply Gradient-based One-Side Sampling (GOSS) to select top gradients and random samples
  Train a decision tree $\Theta'_t(x)$ on the sampled data
  Update the model:
  $\Theta_t(x) = \Theta_{t-1}(x) + \Theta'_t(x)$
end for
return Final model $\hat{\Theta}(x) = \Theta_T(x)$

Algorithm 5 LightGBM Algorithm

- CatBoost

CatBoost, developed by Yandex [212], is a robust open-source ML algorithm known for its extraordinary effectiveness, especially with categorical data. It interacts effortlessly with neural network frameworks such as TensorFlow and CoreML, providing extensive applicability across sectors including healthcare, finance, retail, and natural language processing. CatBoost attains superior accuracy with little data preprocessing by utilising advancements such as effective management of categorical characteristics via target-based encoding and the implementation of Oblivious Decision Trees to reduce overfitting. Although its training duration is marginally extended compared to methods such as GBM or XGBoost, its accelerated prediction time and autonomous handling of missing variables render it exceptionally efficient. This adaptability encompasses various data kinds, including as voice, text, photos, and historical records, rendering it an essential instrument for addressing intricate real-world challenges with precision and simplicity.

$$Z_p(a) = Z_{(p-1)}(a) + \gamma_p l_p(a) \tag{A21}$$

$$\gamma_p = \arg\min \sum_{c=1}^{C} Y(b_c, Z_{(p-1)}(a_c) + \gamma l_p(a_c)) \tag{A22}$$

In Eqs. A21 and A22, the components are defined as follows: the final output after p iterations, representing the cumulative predictions at iteration p. When comparing the actual data to the anticipated data, the loss function calculates the error. The gradients of the loss function in relation to the predictions at iteration p are known as pseudo-residuals. These pseudo residuals guide the model in minimizing the loss function. The number of rounds in the process of improving and the multiplier (or learning rate) for the update at iteration p, this regulates the amount the pseudo-residuals contribute to the finished model.

A.4.3 Stacking

Stacking, also known as stacked generalization is an ensemble learning technique initially introduced by Wolpert in 1992 [213], where it integrates the forecasts of various base learners to construct a more robust model. While bagging and boosting work mostly through aggregation of predictions from the same kind of model, stacking works through aggregating predictions from different types of main models.

Conversely, a collection of different base learners–including neural networks, SVMs, and decision trees–first learn using the same training data in stacking. A more advanced model referred to as the meta-learner or blender utilizes their predictions as inputs to determine the most effective method of combination. Such a two-layer structure might enable a better capture of complex patterns and relationships by itself, which individual models otherwise tend to miss.

Furthermore, the stacking method leverages diverse base algorithms, training them on the identical information to generate numerous models that address the predictive issue from various perspectives. Unlike bagging, which trains decision trees on bootstrapped subsets, this method incorporates several other types of models trained on full data sets. Whereas boosting, which incrementally adjusts model weights to rectify previous errors, stacking determines the optimal approach to combine the predictions of base learners through a meta-model. Similar to logistic regression for classification issues, the meta-model is usually simple because the primary objective of stacking is to combine the outputs of the base models. Linearity is the meta-learner most frequently employed for regression-type problems. [214]. It is this framework that allows stacking to take advantages of strong points from a variety of models to enhance generalization performance. The stacking ensemble algorithm is presenetd in Algorithm 6.

Require: Training dataset $S = \{(x_1, y_1), (x_2, y_2), \ldots, (x_m, y_m)\}$, number of base learners T
Ensure: A stacked ensemble model $H(x)$
Step 1: Train the base learners
for $t = 1$ to T **do**
  Train the t-th base model h_t using dataset S
end for
Step 2: Generate meta-level training data
Create a new dataset where each instance is represented as:
  $\hat{x}_i = \{h_1(x_i), h_2(x_i), \ldots, h_T(x_i)\}$, with target y_i
  This new dataset consists of $\{(\hat{x}_i, y_i)\}_{i=1}^{m}$
Step 3: Train the meta-learner
Train the meta-learner $\hat{h}$ using the newly created dataset
Step 4: Create the final stacked model
Define the final model as:
  $H(x) = \hat{h}(h_1(x), h_2(x), \ldots, h_T(x))$
return $H(x)$

Algorithm 6 Stacking algorithm

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